

3

Motion of a single plasma particle

3.1 Motivation

Single particle motion in neutral gases is trivial – particles move in straight lines until they hit other particles or the wall. Because of this simplicity, there is no point in keeping track of the details of single particle motion in a neutral gas and instead a statistical averaging of this motion suffices; this averaging shows that neutral gases have Maxwellian velocity distributions and are in a local thermodynamic equilibrium. In contrast, plasma particles are nearly collisionless and typically have complex trajectories that are strongly affected by both electric and magnetic fields.

As discussed in the previous chapter, the velocity distribution in a plasma will become Maxwellian when enough collisions have occurred to maximize the entropy. However, since collisions occur infrequently in hot plasmas, many important phenomena have time scales shorter than the time required for the plasma velocity distribution to become Maxwellian. A collisionless model is thus required to characterize these fast phenomena. In these situations randomization does not occur, entropy is conserved, the distribution function need not be Maxwellian, and the plasma is not in thermodynamic equilibrium. Thermodynamic concepts therefore do not apply, and the plasma is instead characterized by concepts from classical mechanics such as momentum or energy conservation of individual particles. In these collisionless situations the complex details of single particle dynamics are not washed out by collisions but instead persist and influence the macroscopic scale. As an example, the cyclotron resonance of a single particle can be important at the macroscopic scale in a collisionless plasma. This chapter examines various aspects of single particle motion and shows how these aspects can influence the macroscopic properties of a plasma.

Furthermore, study of single particle dynamics has a very direct relevance to Vlasov theory because, as shown in Section 2.3, *any* function constructed from constants of single particle motion is a valid solution of the collisionless Vlasov

equation. Thus, knowledge of single particle dynamics provides a “repertoire” of constants of the motion from which solutions to the Vlasov equation suitable for various situations can be constructed.

Finally, the study of single particle motion develops valuable intuition regarding wave–particle interactions and identifies certain unusual situations, such as stochastic or non-adiabatic particle motion, that are beyond the descriptive capability of fluid models.

3.2 Hamilton–Lagrange formalism vs. Lorentz equation

Two mathematically equivalent formalisms describe charged particle dynamics, namely (i) the Lorentz equation

$$m \frac{d\mathbf{v}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (3.1)$$

and (ii) Hamiltonian–Lagrangian dynamics.

The two formalisms are complementary: the Lorentz equation is intuitive and suitable for approximate methods, whereas the more abstract Hamiltonian–Lagrangian formalism exploits time and space symmetries. A brief review of the Hamiltonian–Lagrangian formalism follows, emphasizing aspects relevant to dynamics of charged particles.

The starting point is to postulate the existence of a function L , called the Lagrangian, which:

1. contains *all* information about the particle dynamics for a given situation;
2. depends only on generalized coordinates $Q_i(t)$, $\dot{Q}_i(t)$ appropriate to the problem;
3. possibly has an explicit dependence on time t .

If such a function $L(Q_i(t), \dot{Q}_i(t), t)$ exists, then information on particle dynamics is retrieved by manipulation of the *action integral*

$$S = \int_{t_1}^{t_2} L(Q_i(t), \dot{Q}_i(t), t) dt. \quad (3.2)$$

This manipulation is based on d’Alembert’s principle of least action. According to this principle, one considers the infinity of possible trajectories a particle *could* follow to get from its initial position $Q_i(t_1)$ to its final position $Q_i(t_2)$, and postulates that the trajectory *actually* followed is the one that results in the *lowest* value of S . Thus, the value of S must be minimized (note that S here is action, and not entropy as in the previous chapter). Minimizing S does not give the actual trajectory directly, but rather gives equations of motion, which can be solved to give the actual trajectory.

Minimizing S is accomplished by considering an arbitrary nearby *alternative* trajectory $Q_i(t) + \delta Q_i(t)$ having the *same* beginning and end points as the actual trajectory, i.e., $\delta Q_i(t_1) = \delta Q_i(t_2) = 0$. In order to make the variational argument more precise, δQ_i is expressed as

$$\delta Q_i(t) = \epsilon \eta_i(t), \quad (3.3)$$

where ϵ is an arbitrarily adjustable scalar assumed to be small so that $\epsilon^2 < \epsilon$ and $\eta_i(t)$ is a function of t that vanishes when $t = t_1$ or $t = t_2$ but is otherwise arbitrary. Calculating δS to second order in ϵ gives

$$\begin{aligned} \delta S &= \int_{t_1}^{t_2} L(Q_i + \delta Q_i, \dot{Q}_i + \delta \dot{Q}_i, t) dt - \int_{t_1}^{t_2} L(Q_i, \dot{Q}_i, t) dt \\ &= \int_{t_1}^{t_2} L(Q_i + \epsilon \eta_i, \dot{Q}_i + \epsilon \dot{\eta}_i, t) dt - \int_{t_1}^{t_2} L(Q_i, \dot{Q}_i, t) dt \\ &= \int_{t_1}^{t_2} \left(\epsilon \eta_i \frac{\partial L}{\partial Q_i} + \frac{(\epsilon \eta_i)^2}{2} \frac{\partial^2 L}{\partial Q_i^2} + \epsilon \dot{\eta}_i \frac{\partial L}{\partial \dot{Q}_i} + \frac{(\epsilon \dot{\eta}_i)^2}{2} \frac{\partial^2 L}{\partial \dot{Q}_i^2} \right) dt. \end{aligned} \quad (3.4)$$

Suppose the trajectory $Q_i(t)$ is the one that minimizes S . Any other trajectory must lead to a higher value of S and so δS must be positive for any finite value of ϵ . If ϵ is chosen to be sufficiently small, then the absolute values of the terms of order ϵ^2 in Eq. (3.4) will be smaller than the absolute values of the terms linear in ϵ . The sign of ϵ could then be chosen to make δS negative, but this would violate the requirement that δS must be positive. The only way out of this dilemma is to insist that the sum of the terms linear in ϵ in Eq. (3.4) vanishes so $\delta S \sim \epsilon^2$ and is therefore always positive as required. Insisting that the sum of terms linear in ϵ vanishes implies

$$0 = \int_{t_1}^{t_2} \left(\eta_i \frac{\partial L}{\partial Q_i} + \dot{\eta}_i \frac{\partial L}{\partial \dot{Q}_i} \right) dt. \quad (3.5)$$

Using $\dot{\eta}_i = d\eta_i/dt$ the above expression may be integrated by parts to obtain

$$\begin{aligned} 0 &= \int_{t_1}^{t_2} \left(\eta_i \frac{\partial L}{\partial Q_i} + \frac{d\eta_i}{dt} \frac{\partial L}{\partial \dot{Q}_i} \right) dt \\ &= \left[\eta_i \frac{\partial L}{\partial \dot{Q}_i} \right]_{t_1}^{t_2} + \int_{t_1}^{t_2} \left\{ \eta_i \frac{\partial L}{\partial Q_i} - \eta_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{Q}_i} \right) \right\} dt. \end{aligned} \quad (3.6)$$

Since $\eta_i(t_{1,2}) = 0$, the integrated term vanishes and since η_i was an arbitrary function of t , the coefficient of η_i in the integrand must vanish, yielding *Lagrange's equation*

$$\frac{dP_i}{dt} = \frac{\partial L}{\partial Q_i}, \quad (3.7)$$

where the *canonical momentum* P_i is defined as

$$P_i = \frac{\partial L}{\partial \dot{Q}_i}. \quad (3.8)$$

Equation (3.7) shows that if L does *not* depend on a particular generalized coordinate Q_j , then $dP_j/dt = 0$, in which case the canonical momentum P_j is a *constant of the motion*; the coordinate Q_j is called a *cyclic* or *ignorable* coordinate. This is a very powerful and profound result. Saying that the Lagrangian function does not depend on a coordinate is equivalent to saying that the system is *symmetric* in that coordinate or translationally invariant with respect to that coordinate. The quantities P_j and Q_j are called conjugate and action has the dimensions of the product of these quantities.

Hamilton extended this formalism by introducing a new function related to the Lagrangian. This new function, called the Hamiltonian, provides further useful information and is defined as

$$H \equiv \left(\sum_i P_i \dot{Q}_i \right) - L. \quad (3.9)$$

Partial derivatives of H with respect to P_i and to Q_i give Hamilton's equations

$$\dot{Q}_i = \frac{\partial H}{\partial P_i}, \quad \dot{P}_i = -\frac{\partial H}{\partial Q_i}, \quad (3.10)$$

which are equations of motion having a close relation to phase-space concepts. The time derivative of the Hamiltonian is

$$\frac{dH}{dt} = \sum_i \frac{dP_i}{dt} \dot{Q}_i + \sum_i P_i \frac{d\dot{Q}_i}{dt} - \left(\sum_i \frac{\partial L}{\partial Q_i} \dot{Q}_i + \sum_i \frac{\partial L}{\partial \dot{Q}_i} \frac{d\dot{Q}_i}{dt} + \frac{\partial L}{\partial t} \right) = -\frac{\partial L}{\partial t}. \quad (3.11)$$

This shows that if L does not explicitly depend on time, i.e., $\partial L/\partial t = 0$, the Hamiltonian H is a *constant of the motion*. As will be shown later, H corresponds to the energy of the system, so if $\partial L/\partial t = 0$, the energy is a constant of the motion. Thus, energy is conjugate to time in analogy to canonical momentum being conjugate to position (note that energy $\times$ time also has the units of action). If the Lagrangian does not explicitly depend on time, then the system can be thought of as being “symmetric” with respect to time, or “translationally” invariant with respect to time.

The Lagrangian for a charged particle in an electromagnetic field is

$$L = \frac{mv^2}{2} + q\mathbf{v} \cdot \mathbf{A}(\mathbf{x}, t) - q\phi(\mathbf{x}, t); \quad (3.12)$$

the validity of Eq. (3.12) will now be established by showing that it generates the Lorentz equation when inserted into Lagrange's equation. Since no symmetry is

assumed, there is no reason to use any special coordinate system and so ordinary Cartesian coordinates will be used as the canonical coordinates, in which case Eq. (3.8) gives the canonical momentum as

$$\mathbf{P} = m\mathbf{v} + q\mathbf{A}(\mathbf{x}, t). \quad (3.13)$$

The left-hand side of Eq. (3.7) becomes

$$\frac{d\mathbf{P}}{dt} = m \frac{d\mathbf{v}}{dt} + q \left(\frac{\partial \mathbf{A}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{A} \right), \quad (3.14)$$

while the right-hand side of Eq. (3.7) becomes

$$\begin{aligned} \frac{\partial L}{\partial \mathbf{x}} &= q\nabla(\mathbf{v} \cdot \mathbf{A}) - q\nabla\phi = q(\mathbf{v} \cdot \nabla \mathbf{A} + \mathbf{v} \times \nabla \times \mathbf{A}) - q\nabla\phi \\ &= q(\mathbf{v} \cdot \nabla \mathbf{A} + \mathbf{v} \times \mathbf{B}) - q\nabla\phi. \end{aligned} \quad (3.15)$$

Equating the above two expressions gives the Lorentz equation, where the electric field is defined as $\mathbf{E} = -\partial\mathbf{A}/\partial t - \nabla\phi$ in accordance with Faraday's law. This proves that Eq. (3.12) is mathematically equivalent to the Lorentz equation when used with the principle of least action.

The Hamiltonian associated with this Lagrangian is, in Cartesian coordinates,

$$\begin{aligned} H &= \mathbf{P} \cdot \mathbf{v} - L \\ &= \frac{mv^2}{2} + q\phi \\ &= \frac{(\mathbf{P} - q\mathbf{A}(\mathbf{x}, t))^2}{2m} + q\phi(\mathbf{x}, t), \end{aligned} \quad (3.16)$$

where the last line is the form more suitable for use with Hamilton's equations, i.e., $H = H(\mathbf{x}, \mathbf{P}, t)$. Equation (3.16) also shows that H is, as promised, the particle energy. If generalized coordinates are used, the energy can be written in a general form as $E = H(Q, P, t)$. Equation (3.11) showed that even though both Q and P depend on time, the energy depends on time only if H explicitly depends on time. Thus, in a situation where H does not explicitly depend on time, the energy would have the form $E = H(Q(t), P(t)) = \text{const.}$

It is important to realize that both canonical momentum and energy depend on the reference frame. For example, a bullet fired in an airplane in the direction opposite to the airplane motion, and with a speed equal to the airplane's speed, has a large energy as measured in the airplane frame, but zero energy as measured by an observer on the ground. A more subtle example (of importance to later analysis of waves and Landau damping) occurs when $\mathbf{A}$ and/or ϕ have a wave-like dependence, e.g., $\phi(\mathbf{x}, t) = \phi(\mathbf{x} - \mathbf{v}_{ph}t)$, where $\mathbf{v}_{ph}$ is the wave phase velocity. This potential is *time-dependent* in the lab frame and so the associated Lagrangian

has an explicit dependence on time in the lab frame, which implies that *energy is not a constant of the motion in the lab frame*. In contrast, ϕ is *time-independent* in the wave frame and so the energy is a *constant of the motion in the wave frame*. Existence of a constant of the motion reduces the complexity of the system of equations and typically makes it possible to integrate at least one equation in closed form. Thus, it is advantageous to analyze the system in the frame having the most constants of the motion.

3.3 Adiabatic invariant of a pendulum

Perfect symmetry is never attained in reality. This leads to the practical question of how constants of the motion behave when space and/or time symmetries are “good,” but not perfect. Does the utility of constants of the motion collapse abruptly when the slightest non-symmetrical blemish rears its ugly head, does the utility decay gracefully, or does something completely different happen? To answer these questions, we begin by considering the problem of a small-amplitude pendulum having a time-dependent, but *slowly changing* resonant frequency $\omega(t)$. Since $\omega^2 = g/l$, the time-dependence of the frequency might result from either a slow change in the gravitational acceleration g or else from a slow change in the pendulum length l . In both cases the pendulum equation of motion will be

$$\frac{d^2x}{dt^2} + \omega^2(t)x = 0. \quad (3.17)$$

This equation cannot be solved exactly for arbitrary $\omega(t)$ but if a modest restriction is put on $\omega(t)$ the equation can be solved *approximately* using the WKB method (Wentzel 1926, Kramers 1926, Brillouin 1926). This method is based on the hypothesis that the solution for a time-dependent frequency is likely to be a generalization of the constant-frequency solution

$$x = \text{Re} [A \exp(i\omega t)], \quad (3.18)$$

where this generalization is postulated to be of the form

$$x(t) = \text{Re} \left[A(t) e^{i \int^t \omega(t') dt'} \right]. \quad (3.19)$$

Here $A(t)$ is an amplitude function determined as follows: calculate the first derivative of Eq. (3.19),

$$\frac{dx}{dt} = \text{Re} \left[i\omega A e^{i \int^t \omega(t') dt'} + \frac{dA}{dt} e^{i \int^t \omega(t') dt'} \right], \quad (3.20)$$

then the second derivative

$$\frac{d^2x}{dt^2} = \text{Re} \left[\left(i \frac{d\omega}{dt} A + 2i\omega \frac{dA}{dt} - \omega^2 A + \frac{d^2A}{dt^2} \right) e^{i \int^t \omega(t') dt'} \right], \quad (3.21)$$

and insert this last result into Eq. (3.17) which reduces to

$$i \frac{d\omega}{dt} A + 2i\omega \frac{dA}{dt} + \frac{d^2 A}{dt^2} = 0, \quad (3.22)$$

since the terms involving ω^2 cancel exactly. To proceed further, we make an assumption – the validity of which is to be checked later – that the time dependence of dA/dt is *sufficiently slow* to allow dropping the last term in Eq. (3.22) relative to the middle term. The two terms that remain in Eq. (3.22) can then be rearranged as

$$\frac{1}{\omega} \frac{d\omega}{dt} = -\frac{2}{A} \frac{dA}{dt}, \quad (3.23)$$

which has the exact solution

$$A(t) \sim \frac{1}{\sqrt{\omega(t)}}. \quad (3.24)$$

The assumption of slowness is thus at least self-consistent, for if $\omega(t)$ is indeed slowly changing, Eq. (3.24) shows that $A(t)$ will also be slowly changing and the dropping of the last term in Eq. (3.22) is justified. The slowness requirement can be quantified by assuming that the frequency has an exponential dependence

$$\omega(t) = \omega_0 e^{\alpha t}. \quad (3.25)$$

Thus,

$$\alpha = \frac{1}{\omega} \frac{d\omega}{dt} \quad (3.26)$$

is a measure of how fast the frequency is changing compared to the frequency itself. Hence, dropping the last term in Eq. (3.22) is legitimate if

$$\alpha \ll 4\omega_0 \quad (3.27)$$

or

$$\frac{1}{\omega} \frac{d\omega}{dt} \ll 4\omega. \quad (3.28)$$

In other words, if Eq. (3.28) is satisfied, then the fractional change of the pendulum period per period is small.

Equation (3.24) indicates that when ω is time-dependent, the pendulum amplitude is not constant and so the pendulum energy is *not* conserved. It turns out that what *is* conserved is the *action integral*

$$S = \oint v dx, \quad (3.29)$$

where the integration is over one period of oscillation. This integral can also be written in terms of time as

$$S = \int_{t_0}^{t_0+\tau} v \frac{dx}{dt} dt, \quad (3.30)$$

where t_0 is a time when x is at an instantaneous maximum and τ , the period of a complete cycle, is defined as the interval between two successive times when $dx/dt = 0$ and d^2x/dt^2 has the same sign (e.g., for a pendulum, t_0 would be a time when the pendulum has swung all the way to the right and so is reversing its velocity while τ is the time one has to wait for this to happen again). To show that action is conserved, Eq. (3.29) can be integrated by parts as

$$\begin{aligned} S &= \int_{t_0}^{t_0+\tau} \left[\frac{d}{dt} \left(x \frac{dx}{dt} \right) - x \frac{d^2x}{dt^2} \right] dt \\ &= \left[x \frac{dx}{dt} \right]_{t_0}^{t_0+\tau} - \int_{t_0}^{t_0+\tau} x \frac{d^2x}{dt^2} dt \\ &= \int_{t_0}^{t_0+\tau} \omega^2 x^2 dt, \end{aligned} \quad (3.31)$$

where (i) the integrated term has vanished by virtue of the definitions of t_0 and τ , and (ii) Eq. (3.17) has been used to substitute for d^2x/dt^2 . Equations (3.19) and (3.24) can be combined to give

$$x(t) = x(t_0) \sqrt{\frac{\omega(t_0)}{\omega(t)}} \cos \left(\int_{t_0}^t \omega(t') dt' \right) \quad (3.32)$$

so Eq. (3.31) becomes

$$\begin{aligned} S &= \int_{t_0}^{t_0+\tau} \omega(t')^2 \left\{ x(t_0) \sqrt{\frac{\omega(t_0)}{\omega(t')}} \cos \left(\int_{t_0}^{t'} \omega(t'') dt'' \right) \right\}^2 dt' \\ &= [x(t_0)]^2 \omega(t_0) \int_{t_0}^{t_0+\tau} \omega(t') \cos^2 \left(\int_{t_0}^{t'} \omega(t'') dt'' \right) dt' \\ &= [x(t_0)]^2 \omega(t_0) \int_0^{2\pi} d\xi \cos^2 \xi = \pi [x(t_0)]^2 \omega(t_0) = \text{const.}, \end{aligned} \quad (3.33)$$

where $\xi = \int_{t_0}^{t'} \omega(t'') dt''$ and $d\xi = \omega(t') dt'$. Equation (3.29) shows that S is the area in phase-space enclosed by the trajectory $\{x(t), v(t)\}$ and Eq. (3.33) shows that for a slowly changing pendulum frequency, *this area is a constant of the*

motion. Since the average energy of the pendulum scales as $\sim [\omega(t)x(t)]^2$, we see from Eq. (3.24) that the ratio

$$\frac{\text{energy}}{\text{frequency}} \sim \omega(t)x^2(t) \sim S \sim \text{const.} \quad (3.34)$$

The ratio in Eq. (3.34) is the classical equivalent of the quantum number N of a simple harmonic oscillator because in quantum mechanics the energy E of a simple harmonic oscillator is related to the frequency by the relation $E/\hbar\omega = N + 1/2$.

This analysis clearly applies to *any* dynamical system having an equation of motion of the form of Eq. (3.17). Hence, if the dynamics of plasma particles happens to be of this form, then S can be added to our repertoire of constants of the motion.

3.4 Extension of WKB method to general adiabatic invariant

Action has the dimensions of (canonical momentum) $\times$ (canonical coordinate) so we may anticipate that for general Hamiltonian systems, the action integral given in Eq. (3.29) is not an invariant because v is *not*, in general, proportional to P . We postulate that the general form for the action integral is

$$S = \oint PdQ, \quad (3.35)$$

where the integral is over one period of the periodic motion and P, Q are the relevant canonical momentum–coordinate conjugate pair. The proof of adiabatic invariance used for Eq. (3.29) does not work directly for Eq. (3.35); we now present a slightly more involved proof to show that Eq. (3.35) is indeed the more general form of adiabatic invariant.

Let us define the radius vector in the $Q-P$ plane to be $\mathbf{R} = (Q, P)$ and define unit vectors in the Q and P directions by $\hat{Q}$ and $\hat{P}$; these definitions are shown in Fig. 3.1. Furthermore, we define the z direction as being normal to the $Q-P$

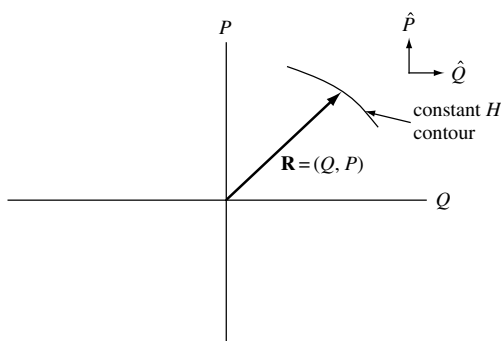


Fig. 3.1 $Q-P$ plane.

plane; thus, the unit vector $\hat{z}$ is “out of the paper,” i.e., $\hat{z} = \hat{Q} \times \hat{P}$. Hamilton’s equations (i.e., $\dot{P} = -\partial H/\partial Q$, $\dot{Q} = \partial H/\partial P$) may be written in vector form as

$$\frac{d\mathbf{R}}{dt} = -\hat{z} \times \nabla H, \quad (3.36)$$

where

$$\nabla = \hat{Q} \frac{\partial}{\partial Q} + \hat{P} \frac{\partial}{\partial P} \quad (3.37)$$

is the gradient operator in the $Q - P$ plane. Equation (3.36) shows that the phase-space “velocity” $d\mathbf{R}/dt$ is *orthogonal* to ∇H . Hence, $\mathbf{R}$ stays on a level contour of H . If H is constant, then, in order for the motion to be periodic, the path along this level contour must circle around and join itself, like a road of *constant elevation* around the rim of a mountain (or a crater). If H is not constant, but slowly changing in time, the contour will circle around and *nearly* join itself.

Equation (3.36) can be inverted by crossing it with $\hat{z}$ to give

$$\nabla H = \hat{z} \times \frac{d\mathbf{R}}{dt}. \quad (3.38)$$

For periodic and near-periodic motions, $d\mathbf{R}/dt$ is always in the same sense (always clockwise or always counterclockwise). Thus, Eq. (3.38) shows that an “observer” following the path would always see that H is increasing on the left-hand side of the path and decreasing on the right-hand side (or vice versa). For clarity, the origin of the $Q - P$ plane is redefined to be at a local maximum or minimum of H . Hence, near the extremum H must have the Taylor expansion

$$H(P, Q) = H_{\text{extremum}} + \frac{P^2}{2} \left[\frac{\partial^2 H}{\partial P^2} \right]_{P=0, Q=0} + \frac{Q^2}{2} \left[\frac{\partial^2 H}{\partial Q^2} \right]_{P=0, Q=0}, \quad (3.39)$$

where $[\partial^2 H/\partial P^2]_{P=0, Q=0}$ and $[\partial^2 H/\partial Q^2]_{P=0, Q=0}$ are either both positive (valley) or both negative (hill). Since H is assumed to have a slow dependence on time, these second derivatives will be time-dependent so that Eq. (3.39) has the form

$$H = \alpha(t) \frac{P^2}{2} + \beta(t) \frac{Q^2}{2}, \quad (3.40)$$

where $\alpha(t)$ and $\beta(t)$ have the same sign. The term H_{extremum} in Eq. (3.39) has been dropped because it is just an additive constant to the energy and does not affect Hamilton’s equations. From Eq. (3.36) the direction of rotation of $\mathbf{R}$ is seen to be counterclockwise if the extremum of H is a hill, and clockwise if a valley.

Hamilton’s equations operating on Eq. (3.40) give

$$\frac{dP}{dt} = -\beta Q, \quad \frac{dQ}{dt} = \alpha P. \quad (3.41)$$

These equations do not directly generate the simple harmonic oscillator equation because of the time dependence of α, β . However, if we define the auxiliary variable

$$\tau = \int^t \beta(t') dt' \quad (3.42)$$

then

$$\frac{d}{dt} = \frac{d\tau}{dt} \frac{d}{d\tau} = \beta \frac{d}{d\tau}$$

so Eq. (3.41) becomes

$$\frac{dP}{d\tau} = -Q, \quad \frac{dQ}{d\tau} = \frac{\alpha}{\beta} P. \quad (3.43)$$

Substituting for Q in the right-hand equation using the left-hand equation gives

$$\frac{d^2 P}{d\tau^2} + \frac{\alpha}{\beta} P = 0, \quad (3.44)$$

which is a simple harmonic oscillator with $\omega^2(\tau) = \alpha(\tau)/\beta(\tau)$. The action integral may be rewritten as

$$S = \int P \frac{dQ}{d\tau} d\tau, \quad (3.45)$$

where the integral is over one period of the motion. Using Eq. (3.43) and following the same procedure as was used with Eq. (3.32) and Eq. (3.33), this becomes

$$S = \int P^2 \frac{\alpha}{\beta} d\tau = \lambda^2 \int \left[\left(\frac{\alpha(\tau')}{\beta(\tau')} \right)^{1/2} \cos^2 \left(\int^{\tau'} (\alpha/\beta)^{1/2} d\tau'' \right) \right] d\tau', \quad (3.46)$$

where λ is a constant dependent on initial conditions. By introducing the orbit phase $\phi = \int^{\tau} (\alpha/\beta)^{1/2} d\tau$, Eq. (3.46) becomes

$$S = \lambda^2 \int_0^{2\pi} d\phi \cos^2 \phi = \text{const.} \quad (3.47)$$

Thus, the general action integral is indeed an adiabatic invariant. This proof is of course only valid in the vicinity of an extremum of H , i.e., only where H can be adequately represented by Eq. (3.40).

3.4.1 Proof for the general adiabatic invariant

We now develop a proof for the general adiabatic invariant. This proof is not restricted to small oscillations (i.e., being near an extremum of H) as was the

previous discussion. Let the Hamiltonian depend on time via a slowly changing parameter $\lambda(t)$, so that $H = H(P, Q, \lambda(t))$. From Eq. (3.16) the energy is given by

$$E(t) = H(P, Q, \lambda(t)) \quad (3.48)$$

and, in principle, this relation can be inverted to give $P = P(E(t), Q, \lambda(t))$. Suppose a particle is executing nearly periodic motion in the $Q - P$ plane. We define the turning point Q_{tp} as a position where $dQ/dt = 0$. Since Q is oscillating, there will be a turning point associated with Q having its maximum value and a turning point associated with Q having its minimum value. From now on let us only consider turning points where Q has its maximum value, that is, we only consider the turning points on the right-hand side of the nearly periodic trajectories in the $Q - P$ plane shown in Fig. 3.2.

If the motion is periodic, then the turning point for the $N + 1^{\text{th}}$ period will be the same as the turning point for the N^{th} period, but if the motion is only nearly periodic, there will be a slight difference as shown in Fig. 3.2. This difference can be characterized by making the turning point a function of time so $Q_{tp} = Q_{tp}(t)$. This function is only defined for the times when $dQ/dt = 0$. When the motion is not exactly periodic, this turning point is such that $Q_{tp}(t + \tau) \neq Q_{tp}(t)$, where τ is the time interval required for the particle to go from the first turning point to the next turning point. The action integral is over one entire period of oscillation starting from a right-hand turning point and then going to the next right-hand turning point (cf. Fig. 3.2) and so can be written as

$$\begin{aligned} S &= \oint P dQ \\ &= \int_{Q_{tp}(t)}^{Q_{tp}(t+\tau)} P dQ. \end{aligned} \quad (3.49)$$

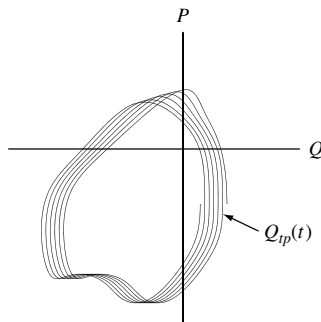


Fig. 3.2 Nearly periodic-phase space trajectory for slowly changing Hamiltonian. The turning point $Q_{tp}(t)$ is where Q is at its maximum.

From Eq. (3.16) it is seen that P/m is not, in general, the velocity and so the velocity dQ/dt is not, in general, proportional to P . Thus, the turning points are not necessarily at the locations where P vanishes, and in fact P need not change sign during a period. However, S still corresponds to the area of phase-space enclosed by one period of the phase-space trajectory.

We can now calculate

$$\begin{aligned}\frac{dS}{dt} &= \frac{d}{dt} \oint P dQ = \frac{d}{dt} \int_{Q_{tp}(t)}^{Q_{tp}(t+\tau)} P(E(t), Q, \lambda(t)) dQ \\ &= \left[P \frac{dQ}{dt} \right]_{Q_{tp}(t)}^{Q_{tp}(t+\tau)} + \int_{Q_{tp}(t)}^{Q_{tp}(t+\tau)} \left(\frac{\partial P}{\partial t} \right)_Q dQ \\ &= \int_{Q_{tp}(t)}^{Q_{tp}(t+\tau)} \left[\left(\frac{\partial P}{\partial E} \right)_{Q,\lambda} \frac{dE}{dt} + \left(\frac{\partial P}{\partial \lambda} \right)_{Q,E} \frac{d\lambda}{dt} \right] dQ.\end{aligned}\quad (3.50)$$

Because $dQ/dt = 0$ at the turning point, the integrated term vanishes and so there is no contribution from motion of the turning point. From Eq. (3.48) we have

$$1 = \frac{\partial H}{\partial P} \left(\frac{\partial P}{\partial E} \right)_{Q,\lambda} \quad (3.51)$$

and

$$0 = \frac{\partial H}{\partial P} \left(\frac{\partial P}{\partial \lambda} \right)_{Q,E} + \frac{\partial H}{\partial \lambda} \quad (3.52)$$

so Eq. (3.50) becomes

$$\frac{dS}{dt} = \oint \left(\frac{\partial H}{\partial P} \right)^{-1} \left[\frac{dE}{dt} - \frac{\partial H}{\partial \lambda} \frac{d\lambda}{dt} \right] dQ. \quad (3.53)$$

From Eq. (3.48) we have

$$\frac{dE}{dt} = \frac{\partial H}{\partial P} \frac{dP}{dt} + \frac{\partial H}{\partial Q} \frac{dQ}{dt} + \frac{\partial H}{\partial \lambda} \frac{d\lambda}{dt} = \frac{\partial H}{\partial \lambda} \frac{d\lambda}{dt}, \quad (3.54)$$

since the first two terms canceled due to Hamilton's equations. Substitution of Eq. (3.54) into Eq. (3.53) gives $dS/dt = 0$, completing the proof of adiabatic invariance. No assumption has been made here that P, Q are close to the values associated with an extremum of H .

This proof seems too neat, because it has established adiabatic invariance simply by careful use of the chain rule, and by taking partial derivatives. However, this observation reveals the underlying essence of adiabaticity, namely it is the differentiability of H, P with respect to λ from one period to the next and the

Hamilton nature of the system, which together provide the conditions for the adiabatic invariant to exist. If the motion had been such that after one cycle the motion had changed so drastically that taking a derivative of H or P with respect to λ would not make sense, then the adiabatic invariant would not exist.

3.5 Drift equations

We show in this section that it is possible to deduce intuitive and quite accurate analytic solutions for the velocity (drift) of charged particles in arbitrarily complicated electric and magnetic fields provided the fields are *slowly changing in both space and time* (this requirement is essentially the slowness requirement for adiabatic invariance). Drift solutions are obtained by solving the Lorentz equation

$$m \frac{d\mathbf{v}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (3.55)$$

iteratively, taking advantage of the assumed separation of scales between fast and slow motions.

3.5.1 Simple $\mathbf{E} \times \mathbf{B}$ and force drifts

Before developing the general method for analyzing drifts, a simple example illustrating the basic idea will now be discussed. This example consists of an ion starting at rest in a spatially uniform magnetic field $\mathbf{B} = B\hat{z}$ and a spatially uniform electric field $\mathbf{E} = E\hat{y}$. The origin is defined to be at the ion's starting position and both electric and magnetic fields are constant in time. The assumed spatial uniformity and time-independence of the fields represent the extreme limit of assuming that the fields are slowly changing in space and time.

Because the magnetic force $q\mathbf{v} \times \mathbf{B}$ is perpendicular to $\mathbf{v}$, the magnetic force does no work and so only the electric field can change the ion's energy (this can be seen by dotting Eq. (3.55) with $\mathbf{v}$). Also, because all fields are uniform and static the electric field can be expressed as $\mathbf{E} = -\nabla\phi$, where $\phi = -Ey$ is an electrostatic potential. Since the ion lowers its potential energy $q\phi$ on moving to larger y , motion in the positive y direction corresponds to the ion "falling downhill." Since the ion starts from rest at $y = 0$, where $\phi = 0$, its total energy $W = mv^2/2 + q\phi$ is initially zero. Furthermore, the time-independence of the fields implies that W must remain zero for all time. Because the kinetic energy $mv^2/2$ is positive-definite, the ion can only attain finite kinetic energy if it falls downhill, i.e., moves into regions of positive y . If for any reason the ion y -coordinate becomes zero at some later time, then at such a time the ion would again have to have $\mathbf{v} = 0$ because $W = mv^2 - qEy = 0$.

When the ion begins moving, it is acted on primarily by the electric force $qE\hat{y}$ because the magnetic force $q\mathbf{v} \times \mathbf{B}$ is negligible at small velocities. The electric force accelerates the ion in the y direction so the ion develops a positive v_y and also moves towards larger positive y as it “falls downhill” in the potential. As it develops a positive v_y , the ion starts to experience a magnetic force $qv_y\hat{y} \times B\hat{z} = v_y qB\hat{x}$, which accelerates the ion in the positive x direction causing the ion to develop in addition a positive v_x . The trajectory now becomes curved as the ion veers in the x direction while moving towards larger y . The positive v_x continues to increase and as a consequence a new magnetic force $qv_x\hat{x} \times B\hat{z} = -v_x qB\hat{y}$ develops and, being in the negative y direction, this increasing magnetic force counteracts the steady electric force, eventually causing the ion to decelerate in the y direction. The velocity v_y now decreases and ultimately reverses so that the ion starts to head in the negative y direction back towards $y = 0$. As a consequence of the reversal of v_y , the magnetic force $qv_y\hat{y} \times B\hat{z}$ will become negative and so the ion will also decelerate in the x direction. Moving with negative v_y means the ion is going uphill in the electrostatic potential and when it reaches $y = 0$, its potential energy must go back to zero. As noted above, the ion must come to rest at this point, because its total energy is always zero. Because the x velocity was never negative, the result of all this is that the ion makes a net positive displacement in the x direction. The whole process then repeats with the result that the ion keeps advancing in x while making a sequence of semi-circles in which v_y oscillates in polarity while v_x is never negative. The ion consequently moves like a leap-frog, which bounces up and down in the y direction while continuously advancing in the x direction. If an electron had been used instead of an ion, the sign of both the electric and magnetic forces would have reversed and the electron would have been confined to regions where $y \leq 0$. However, the net displacement would also be in the positive x direction (this is easily seen by repeating the above argument using an electron).

If an ion starts with a finite rather than a zero velocity, it will execute cyclotron (also called Larmor) orbits, which take the ion into regions of both positive and negative y . However, the ion will have a larger gyroradius in its $y > 0$ orbit segment than in its $y < 0$ orbit segment, resulting again in an average drift to the right as shown in Fig. 3.3. Electrons have larger gyroradii in the $y < 0$ portions of their orbit, but have a counterclockwise rotation so electrons *also* drift to the right. The magnitude of this steady drift is easily calculated by assuming the existence of a constant perpendicular drift velocity in the Lorentz equation, and then averaging out the cyclotron motion:

$$0 = \mathbf{E} + \langle \mathbf{v} \rangle \times \mathbf{B}. \quad (3.56)$$

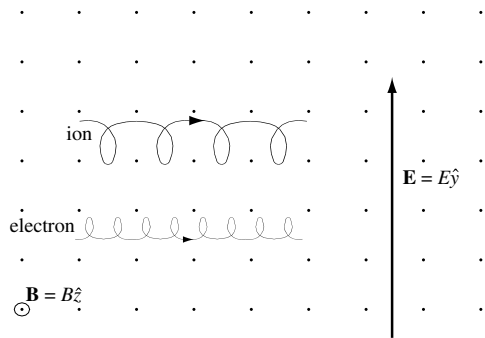


Fig. 3.3 $\mathbf{E} \times \mathbf{B}$ drifts for particles having finite initial energy.

This may be solved to give the average drift velocity

$$\mathbf{v}_E = \langle \mathbf{v} \rangle = \frac{\mathbf{E} \times \mathbf{B}}{B^2}. \tag{3.57}$$

This steady drift, called the $\mathbf{E} \times \mathbf{B}$ drift (pronounced “E cross B”), is independent of both the particle’s polarity and its initial velocity. One way of interpreting this behavior is to recall that according to the theory of special relativity the electric field $\mathbf{E}'$ observed in a frame moving with velocity $\mathbf{u}$ is $\mathbf{E}' = \mathbf{E} + \mathbf{u} \times \mathbf{B}$ and so Eq. (3.56) is simply a statement that a particle drifts in such a way to ensure that the electric field seen in its own frame vanishes. The $\mathbf{E} \times \mathbf{B}$ drift analysis can be easily generalized to describe the effect on a charged particle of *any* force orthogonal to $\mathbf{B}$ by simply making the replacement $\mathbf{E} \rightarrow \mathbf{F}/q$ in the Lorentz equation. Thus, any spatially uniform, temporally constant force orthogonal to $\mathbf{B}$ will cause a drift

$$\mathbf{v}_F \equiv \langle \mathbf{v} \rangle = \frac{\mathbf{F} \times \mathbf{B}}{qB^2}. \tag{3.58}$$

Equations (3.57) and (3.58) lead to two counter-intuitive and important conclusions:

- 1. A steady-state electric field perpendicular to a magnetic field does not drive currents in a plasma, but instead causes a bulk motion of the entire plasma across the magnetic field with the velocity $\mathbf{v}_E$.
- 2. A steady-state *force* (e.g., gravity, centrifugal force, etc.) perpendicular to the magnetic field causes oppositely directed motions for electrons and ions and so drives a cross-field current

$$\mathbf{J}_F = \sum_{\sigma} n_{\sigma} \frac{\mathbf{F} \times \mathbf{B}}{B^2}. \tag{3.59}$$

3.5.2 Drifts in slowly changing arbitrary fields

We now consider charged particle motion in arbitrarily complicated but slowly changing fields subject to the following restrictions:

1. The time variation is so slow that the fields can be considered as approximately constant during each cyclotron period of the motion.
2. The fields vary so *gradually* in space that they are nearly uniform over the spatial extent of any single complete cyclotron orbit.
3. The electric and magnetic fields are related by Faraday's law $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$.
4. $E/B \ll c$ so that relativistic effects are unimportant (otherwise there would be a problem with v_E becoming faster than c).

In this more general situation a charged particle will gyrate about $\mathbf{B}$, stream parallel to $\mathbf{B}$, have $\mathbf{E} \times \mathbf{B}$ drifts across $\mathbf{B}$, and may also have force-based drifts. The analysis is based on the assumption that all these various motions are well separated (i.e., easily distinguishable from each other); this assumption is closely related to the requirement that the fields vary slowly and also to the concept of adiabatic invariance.

The assumed separation of scales is expressed by decomposing the particle motion into a fast, oscillatory component – the gyromotion – and a slow component obtained by averaging out the gyromotion. As sketched in Fig. 3.4, the particle's position and velocity are each decomposed into two terms

$$\mathbf{x}(t) = \mathbf{x}_{gc}(t) + \mathbf{r}_L(t), \quad \mathbf{v}(t) = \frac{d\mathbf{x}}{dt} = \mathbf{v}_{gc}(t) + \mathbf{v}_L(t), \quad (3.60)$$

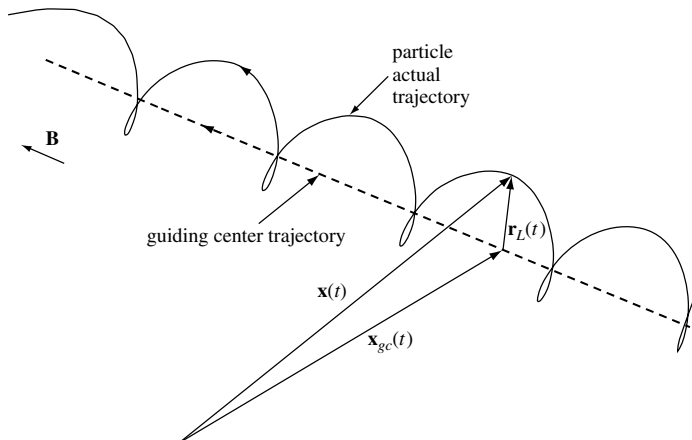


Fig. 3.4 Drift in an arbitrarily complicated field.

where $\mathbf{r}_L(t)$, $\mathbf{v}_L(t)$ give the *fast* gyration of the particle in a cyclotron orbit and $\mathbf{x}_{gc}(t)$, $\mathbf{v}_{gc}(t)$ are the *slowly* changing motion of the *guiding center* obtained after averaging out the cyclotron motion. Ignoring any time dependence of the fields for now, the magnetic field seen by the particle can be written as

$$\begin{aligned}\mathbf{B}(\mathbf{x}(t)) &= \mathbf{B}(\mathbf{x}_{gc}(t) + \mathbf{r}_L(t)) \\ &= \mathbf{B}(\mathbf{x}_{gc}(t)) + (\mathbf{r}_L(t) \cdot \nabla) \mathbf{B}.\end{aligned}\quad (3.61)$$

Because $\mathbf{B}$ was assumed to be nearly uniform over the cyclotron orbit, it is sufficient to keep only the first term in the Taylor expansion of the magnetic field. The electric field may be expanded in a similar fashion.

After insertion of these Taylor expansions for the non-uniform electric and magnetic fields, the Lorentz equation becomes

$$\begin{aligned}m \frac{d[\mathbf{v}_{gc}(t) + \mathbf{v}_L(t)]}{dt} &= q [\mathbf{E}(\mathbf{x}_{gc}(t)) + (\mathbf{r}_L(t) \cdot \nabla) \mathbf{E}] \\ &\quad + q [\mathbf{v}_{gc}(t) + \mathbf{v}_L(t)] \times [\mathbf{B}(\mathbf{x}_{gc}(t)) + (\mathbf{r}_L(t) \cdot \nabla) \mathbf{B}].\end{aligned}\quad (3.62)$$

The gyromotion (i.e., the fast cyclotron motion) is defined to be the solution of the equation

$$m \frac{d\mathbf{v}_L(t)}{dt} = q \mathbf{v}_L(t) \times \mathbf{B}(\mathbf{x}_{gc}(t)); \quad (3.63)$$

subtracting this fast motion equation from Eq. (3.62) leaves

$$\begin{aligned}m \frac{d\mathbf{v}_{gc}(t)}{dt} &= q [\mathbf{E}(\mathbf{x}_{gc}(t)) + (\mathbf{r}_L(t) \cdot \nabla) \mathbf{E}] \\ &\quad + q \{ \mathbf{v}_{gc}(t) \times [\mathbf{B}(\mathbf{x}_{gc}(t)) + (\mathbf{r}_L(t) \cdot \nabla) \mathbf{B}] + \mathbf{v}_L(t) \times (\mathbf{r}_L(t) \cdot \nabla) \mathbf{B} \}.\end{aligned}\quad (3.64)$$

Let us now average Eq. (3.64) over one gyroperiod in which case terms *linear* in gyromotion average to zero. What remains is an equation describing the slow quantities, namely

$$m \frac{d\mathbf{v}_{gc}(t)}{dt} = q \{ \mathbf{E}(\mathbf{x}_{gc}(t)) + \mathbf{v}_{gc}(t) \times \mathbf{B}(\mathbf{x}_{gc}(t)) + \langle \mathbf{v}_L(t) \times (\mathbf{r}_L(t) \cdot \nabla) \mathbf{B} \rangle \}, \quad (3.65)$$

where $\langle \rangle$ means averaged over a cyclotron period. The guiding center velocity can now be decomposed into components perpendicular and parallel to $\mathbf{B}$,

$$\mathbf{v}_{gc}(t) = \mathbf{v}_{\perp gc}(t) + v_{\parallel gc}(t) \hat{\mathbf{B}} \quad (3.66)$$

so that

$$\frac{d\mathbf{v}_{gc}(t)}{dt} = \frac{d\mathbf{v}_{\perp gc}(t)}{dt} + \frac{d(v_{\parallel gc}(t) \hat{\mathbf{B}})}{dt} = \frac{d\mathbf{v}_{\perp gc}(t)}{dt} + \frac{dv_{\parallel gc}(t)}{dt} \hat{\mathbf{B}} + v_{\parallel gc}(t) \frac{d\hat{\mathbf{B}}}{dt}. \quad (3.67)$$

Denoting the distance along the magnetic field by s , the derivative of the magnetic field unit vector can be written, to lowest order, as

$$\frac{d\hat{\mathbf{B}}}{dt} = \frac{\partial \hat{\mathbf{B}}}{\partial s} \frac{ds}{dt} = v_{\parallel gc} \hat{\mathbf{B}} \cdot \nabla \hat{\mathbf{B}}, \quad (3.68)$$

so Eq. (3.65) becomes

$$\begin{aligned} m \left[\frac{d\mathbf{v}_{\perp gc}(t)}{dt} + \frac{dv_{\parallel gc}(t)}{dt} \hat{\mathbf{B}} + v_{\parallel gc}^2 \hat{\mathbf{B}} \cdot \nabla \hat{\mathbf{B}} \right] &= q\mathbf{E}(\mathbf{x}_{gc}(t)) \\ &+ q\mathbf{v}_{gc}(t) \times \mathbf{B}(\mathbf{x}_{gc}(t)) \\ &+ q \langle \mathbf{v}_L(t) \times (\mathbf{r}_L(t) \cdot \nabla) \mathbf{B} \rangle. \end{aligned} \quad (3.69)$$

The component of this equation along $\mathbf{B}$ is

$$m \frac{dv_{\parallel gc}(t)}{dt} = q \left[E_{\parallel}(\mathbf{x}_{gc}(t)) + \langle \mathbf{v}_L(t) \times (\mathbf{r}_L(t) \cdot \nabla) \mathbf{B} \rangle_{\parallel} \right] \quad (3.70)$$

while the component perpendicular to $\mathbf{B}$ is

$$m \left[\frac{d\mathbf{v}_{\perp gc}(t)}{dt} + v_{\parallel gc}^2 \hat{\mathbf{B}} \cdot \nabla \hat{\mathbf{B}} \right] = q \left[\begin{aligned} &\mathbf{E}_{\perp}(\mathbf{x}_{gc}(t)) \\ &+ \mathbf{v}_{gc}(t) \times \mathbf{B}(\mathbf{x}_{gc}(t)) \\ &+ \langle \mathbf{v}_L(t) \times (\mathbf{r}_L(t) \cdot \nabla) \mathbf{B} \rangle_{\perp} \end{aligned} \right]. \quad (3.71)$$

Equation (3.71) is of the generic form

$$m \frac{d\mathbf{v}_{\perp gc}}{dt} = \mathbf{F}_{\perp} + q\mathbf{v}_{gc} \times \mathbf{B}. \quad (3.72)$$

where

$$\begin{aligned} \mathbf{F}_{\perp} &= q \left[\mathbf{E}_{\perp}(\mathbf{x}_{gc}(t)) + \langle \mathbf{v}_L(t) \times (\mathbf{r}_L(t) \cdot \nabla) \mathbf{B} \rangle_{\perp} \right] \\ &- m v_{\parallel gc}^2 \hat{\mathbf{B}} \cdot \nabla \hat{\mathbf{B}}. \end{aligned} \quad (3.73)$$

Equation (3.72) is solved iteratively based on the assumption that $\mathbf{v}_{\perp gc}$ has a slow time dependence. In the first iteration, the time dependence is neglected altogether so that the left-hand side of Eq. (3.72) is set to zero to obtain the “first guess” for the perpendicular drift to be

$$\mathbf{v}_{\perp gc} \simeq \mathbf{v}_F \equiv \frac{\mathbf{F}_{\perp} \times \mathbf{B}}{qB^2}.$$

Next, $\mathbf{v}_p$ is defined to be a correction to this first guess, where $\mathbf{v}_p$ is assumed small and incorporates effects due to any time dependence of $\mathbf{v}_{\perp gc}$. To determine $\mathbf{v}_p$, we write $\mathbf{v}_{\perp gc} = \mathbf{v}_F + \mathbf{v}_p$ so, to second order Eq. (3.72) becomes

$$m \frac{d(\mathbf{v}_F + \mathbf{v}_p)}{dt} = \mathbf{F}_{\perp} + q(\mathbf{v}_F + \mathbf{v}_p) \times \mathbf{B}. \quad (3.74)$$

In accordance with the slowness condition, it is assumed that $|\mathbf{dv}_p/\mathbf{dt}| \ll |\mathbf{dv}_F/\mathbf{dt}|$ so Eq. (3.74) becomes

$$0 = -m \frac{d\mathbf{v}_F}{dt} + q\mathbf{v}_p \times \mathbf{B}. \quad (3.75)$$

Crossing this equation with $\mathbf{B}$ gives the general *polarization drift*

$$\mathbf{v}_p = -\frac{m}{qB^2} \frac{d\mathbf{v}_F}{dt} \times \mathbf{B}. \quad (3.76)$$

The most important example of the polarization drift is when $\mathbf{v}_F$ is the $\mathbf{E} \times \mathbf{B}$ drift in a uniform, constant magnetic field so that

$$\begin{aligned} \mathbf{v}_p &= -\frac{m}{qB^2} \frac{d}{dt} \left(\frac{\mathbf{E} \times \mathbf{B}}{B^2} \right) \times \mathbf{B} \\ &= \frac{m}{qB^2} \frac{d\mathbf{E}}{dt}. \end{aligned} \quad (3.77)$$

We now evaluate the middle term in Eq. (3.73); this term is called the “grad B ” force and is

$$\mathbf{F}_{\nabla B} = q \langle \mathbf{v}_L(t) \times (\mathbf{r}_L(t) \cdot \nabla) \mathbf{B} \rangle. \quad (3.78)$$

To simplify the algebra for the averaging over cyclotron orbits, a local Cartesian coordinate system is used with x axis in the direction of the gyrovelocity at $t = 0$ and z axis in the direction of the magnetic field *at the gyrocenter*. Thus, the Larmor orbit velocity has the form

$$\mathbf{v}_L(t) = v_L [\hat{x} \cos \omega_c t - \hat{y} \sin \omega_c t], \quad (3.79)$$

where

$$\omega_c = \frac{qB}{m} \quad (3.80)$$

is the cyclotron frequency and the Larmor orbit position has the form

$$\mathbf{r}_L(t) = \frac{v_L}{\omega_c} [\hat{x} \sin \omega_c t + \hat{y} \cos \omega_c t]. \quad (3.81)$$

Inserting the above two expressions in Eq. (3.78) gives

$$\mathbf{F}_{\nabla B} = q \frac{v_L^2}{\omega_c} \langle [\hat{x} \cos \omega_c t - \hat{y} \sin \omega_c t] \times ([\hat{x} \sin \omega_c t + \hat{y} \cos \omega_c t] \cdot \nabla) \mathbf{B} \rangle. \quad (3.82)$$

Noting that $\langle \sin^2 \omega_c t \rangle = \langle \cos^2 \omega_c t \rangle = 1/2$ while $\langle \sin(\omega_c t) \cos(\omega_c t) \rangle = 0$, this reduces to

$$\begin{aligned} \mathbf{F}_{\nabla B} &= \frac{qv_L^2}{2\omega_c} \left[\hat{x} \times \frac{\partial \mathbf{B}}{\partial y} - \hat{y} \times \frac{\partial \mathbf{B}}{\partial x} \right] \\ &= \frac{mv_L^2}{2B} \left[\hat{x} \times \frac{\partial (B_y \hat{y} + B_z \hat{z})}{\partial y} - \hat{y} \times \frac{\partial (B_x \hat{x} + B_z \hat{z})}{\partial x} \right] \\ &= \frac{mv_L^2}{2B} \left[\hat{z} \left(\frac{\partial B_y}{\partial y} + \frac{\partial B_x}{\partial x} \right) - \hat{y} \frac{\partial B_z}{\partial y} - \hat{x} \frac{\partial B_z}{\partial x} \right]. \end{aligned} \quad (3.83)$$

But from $\nabla \cdot \mathbf{B} = 0$, it is seen that $\frac{\partial B_y}{\partial y} + \frac{\partial B_x}{\partial x} = -\frac{\partial B_z}{\partial z}$, so the “grad B ” force is

$$\mathbf{F}_{\nabla B} = -\frac{mv_L^2}{2B} \nabla B, \quad (3.84)$$

where the approximation $B_z \simeq B$ has been used, since the magnetic field direction is mainly in the $\hat{z}$ direction.

Let us now define

$$\mathbf{F}_c = -mv_{\parallel gc}^2 \hat{B} \cdot \nabla \hat{B} \quad (3.85)$$

and consider this force. Suppose the magnetic field lines are curved and consider a particular point on a specific field line. Define, as shown in Fig. 3.5, a two-dimensional cylindrical coordinate system (R, ϕ) with origin at the field line

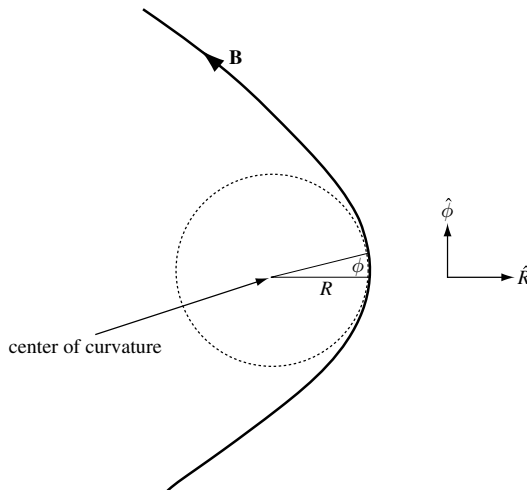


Fig. 3.5 Local cylindrical coordinate system defined by curved magnetic field, $\hat{\phi} = \hat{B}$.

center of curvature for this specific point and lying in the plane of the field line at this point. The radial position of the chosen point in this cylindrical coordinate system is the local radius of curvature of the field line and, since $\hat{\phi} = \hat{B}$, it is seen that $\hat{B} \cdot \nabla \hat{B} = \hat{\phi} \cdot \nabla \hat{\phi} = -\hat{R}/R$. The force associated with curvature of a field line,

$$\mathbf{F}_c = \frac{mv_{\parallel gc}^2 \hat{R}}{R}, \quad (3.86)$$

is therefore just the centrifugal force resulting from the motion along the curve of the particle's guiding center.

The drifts can be summarized as

$$\mathbf{v}_{\perp gc} = \mathbf{v}_E + \mathbf{v}_{\nabla B} + \mathbf{v}_c + \mathbf{v}_p, \quad (3.87)$$

where:

1. the “ $\mathbf{E}$ cross $\mathbf{B}$ ” drift is

$$\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2}; \quad (3.88)$$

2. the “grad B ” drift is

$$\mathbf{v}_{\nabla B} = -\frac{mv_L^2}{2qB^3} \nabla B \times \mathbf{B}; \quad (3.89)$$

3. the “curvature” drift is

$$\mathbf{v}_c = -\frac{mv_{\parallel gc}^2}{qB^2} \hat{B} \cdot \nabla \hat{B} \times \mathbf{B} = \frac{1}{qB^2} \left(\frac{mv_{\parallel gc}^2 \hat{R}}{R} \right) \times \mathbf{B}; \quad (3.90)$$

4. the “polarization” drift is

$$\mathbf{v}_p = -\frac{m}{qB^2} \left[\frac{d}{dt} (\mathbf{v}_E + \mathbf{v}_{\nabla B} + \mathbf{v}_c) \right] \times \mathbf{B}. \quad (3.91)$$

3.5.3 μ conservation

We now imagine being in a frame moving with the velocity $\mathbf{v}_{\perp gc}$; in this frame the only perpendicular velocity is the cyclotron velocity (Larmor motion). Since $\mathbf{v}_{\perp gc}$ is orthogonal to $\mathbf{B}$, the parallel equation of motion is not affected by this change of frame and using Eqs. (3.70) and (3.84) can be written as

$$m \frac{dv_{\parallel}}{dt} = qE_{\parallel} - \frac{mv_L^2}{2B} \frac{\partial B}{\partial s}, \quad (3.92)$$

where, as before, s is the distance along the magnetic field. Multiplication by $v_{\parallel}$ gives an energy relation

$$\frac{d}{dt} \left(\frac{mv_{\parallel}^2}{2} \right) = qE_{\parallel} v_{\parallel} - \frac{mv_L^2}{2B} v_{\parallel} \frac{\partial B}{\partial s}. \quad (3.93)$$

The perpendicular force defined in Eq. (3.73) does not exist in this moving frame because it has been “transformed away” by the change of frames. Also, recall that it was assumed that the characteristic scale lengths of $\mathbf{E}$ and $\mathbf{B}$ are large compared to the gyroradius (Larmor radius). However, if the magnetic field has an *absolute time derivative*, Faraday’s law dictates that there must be an inductive electric field, i.e., an electric field where $\oint \mathbf{E} \cdot d\mathbf{l}$ is finite. Because $\oint \mathbf{E} \cdot d\mathbf{l} \neq 0$, the force provided by an inductive field is *non-conservative* and so is intrinsically different from the conservative force provided by the static electric field previously discussed. The consequences of this inductive field and its non-conservative force must therefore be taken into account explicitly.

To understand the effect of an inductive electric field on a specific particle, we dot the Lorentz equation with $\mathbf{v}$ to obtain

$$\frac{d}{dt} \left(\frac{mv_{\parallel}^2}{2} + \frac{mv_L^2}{2} \right) = qv_{\parallel} E_{\parallel} + q\mathbf{v}_{\perp} \cdot \mathbf{E}_{\perp}, \quad (3.94)$$

where $\mathbf{v}_{\perp}$ is the vector Larmor orbit velocity and $v_L = \sqrt{\mathbf{v}_{\perp} \cdot \mathbf{v}_{\perp}}$ is its scalar magnitude. Subtracting Eq. (3.93) from (3.94) gives

$$\frac{d}{dt} \left(\frac{mv_L^2}{2} \right) = q\mathbf{v}_{\perp} \cdot \mathbf{E}_{\perp} + \frac{mv_L^2}{2B} v_{\parallel} \frac{\partial B}{\partial s}. \quad (3.95)$$

Integration of Faraday’s law over the cross-section of the Larmor orbit gives

$$\int d\mathbf{s} \cdot \nabla \times \mathbf{E} = - \int d\mathbf{s} \cdot \frac{\partial \mathbf{B}}{\partial t} \quad (3.96)$$

or

$$\oint d\mathbf{l} \cdot \mathbf{E} = -\pi r_L^2 \frac{\partial B}{\partial t}, \quad (3.97)$$

where it has been assumed that the magnetic field is changing sufficiently slowly for the orbit radius to be approximately constant during each orbit.

Equation (3.95) involves the local electric field $\mathbf{E}_{\perp}$ but Eq. (3.97) only gives the line integral of the electric field. This line integral can still be used if Eq. (3.95) is averaged over a cyclotron period. The critical term is the time average over the

Larmor orbit of $q\mathbf{v}_\perp \cdot \mathbf{E}_\perp$ (which gives the rate at which the perpendicular electric field does work on the particle),

$$\begin{aligned} \langle q\mathbf{v}_\perp \cdot \mathbf{E}_\perp \rangle_{orbit} &= \frac{\omega_c}{2\pi} \int dt \, q\mathbf{v}_\perp \cdot \mathbf{E}_\perp \\ &= -\frac{q\omega_c}{2\pi} \oint d\mathbf{l} \cdot \mathbf{E}_\perp \\ &= \frac{q\omega_c}{2} r_L^2 \frac{\partial B}{\partial t}. \end{aligned} \quad (3.98)$$

The substitution $\mathbf{v}_\perp dt = -d\mathbf{l}$ has been used and the minus sign is invoked because particle motion is diamagnetic (e.g., ions have a left-handed orbit, whereas in Stokes' theorem $d\mathbf{l}$ is assumed to be a right-handed line element). Averaging of Eq. (3.95) gives

$$\left\langle \frac{d}{dt} \left(\frac{mv_L^2}{2} \right) \right\rangle = \frac{mv_L^2}{2B} \frac{\partial B}{\partial t} + \frac{mv_L^2}{2B} v_\parallel \frac{\partial B}{\partial s} = \frac{mv_L^2}{2B} \frac{dB}{dt}, \quad (3.99)$$

where $dB/dt = \partial B/\partial t + v_\parallel \partial B/\partial s$ is the *total derivative* of the average magnetic field experienced by the particle over a Larmor orbit. Defining the Larmor orbit kinetic energy as $W_\perp = mv_L^2/2$, Eq. (3.99) can be rewritten as

$$\frac{1}{W_\perp} \frac{dW_\perp}{dt} = \frac{1}{B} \frac{dB}{dt}, \quad (3.100)$$

which has the solution

$$\frac{W_\perp}{B} \equiv \mu = \text{const.} \quad (3.101)$$

for magnetic fields that can be changing in *both* time and space. In plasma physics terminology, μ is called the “first adiabatic” invariant, and the invariance of μ shows that the ratio of the kinetic energy of gyromotion to gyrofrequency is a conserved quantity. The derivation assumed the magnetic field changed sufficiently slowly for the instantaneous field strength $B(t)$ during an orbit to differ only slightly from the orbit-averaged field strength $\langle B \rangle$, i.e., $|B(t) - \langle B \rangle| \ll \langle B \rangle$.

3.5.4 Relation of μ conservation to other conservation relations

μ conservation is both of fundamental importance and a prime example of the adiabatic invariance of the action integral associated with a periodic motion. The μ conservation concept unites several seemingly disparate points of view, namely:

1. *Conservation of magnetic moment of a particle* – According to electromagnetic theory¹ the magnetic moment m of a current loop is $m = IA$, where I is the current flowing in

¹ For example, see p. 186 of (Jackson 1998).

the loop and A is the area enclosed by the loop. Because a gyrating particle traces out a circular orbit at the frequency $\omega_c/2\pi$ and has a charge q , it effectively constitutes a current loop having $I = q\omega_c/2\pi$ and cross-sectional area $A = \pi r_L^2$. Thus, the magnetic moment of the gyrating particle is

$$m = \left(\frac{q\omega_c}{2\pi} \right) \pi r_L^2 = \frac{mv_L^2}{2B} = \mu \quad (3.102)$$

and so the magnetic moment m is an adiabatically conserved quantity.

2. *Conservation of magnetic flux enclosed by gyro-orbit* – Because the magnetic flux Φ enclosed by the gyro-orbit is

$$\Phi = B\pi r_L^2 = \left(\frac{2m\pi}{q^2} \right) \mu, \quad (3.103)$$

μ conservation further implies conservation of the magnetic flux enclosed by a gyro-orbit. This is consistent with the concept that magnetic flux is frozen into the plasma, since if the field is made stronger, the field lines squeeze together in such a way that the density of field lines per unit area increases in proportion to the field strength. As shown in Fig. 3.6, the particle orbit area contracts in inverse proportion to the field strength so after a compression of field, the particle orbit links the same number of field lines as before the compression.

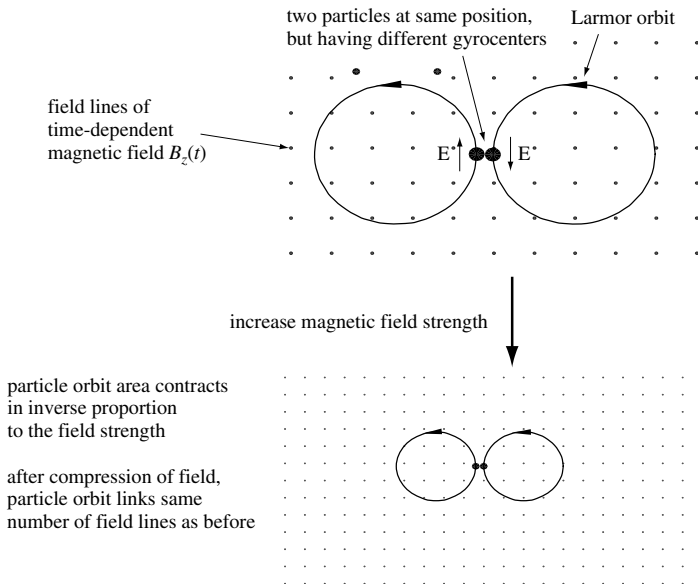


Fig. 3.6 Illustration showing how conservation of flux linked by an orbit is equivalent to frozen-in field; also, increasing magnetic field results in magnetic compression.

3. *Hamiltonian point of view (cylindrical geometry with azimuthal symmetry)* – Define a cylindrical coordinate system (r, θ, z) with z axis along the axis of rotation of the gyrating particle. Since $B_z = r^{-1} \partial(rA_\theta)/\partial r$, the vector potential is $A_\theta = rB_z/2$. The velocity vector is $\mathbf{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + \dot{z}\hat{z}$ and the Lagrangian is

$$L = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2) + qr\dot{\theta}A_\theta - q\phi \quad (3.104)$$

so the canonical angular momentum is

$$P_\theta = mr^2\dot{\theta} + qrA_\theta = mr^2\dot{\theta} + qr^2B_z/2. \quad (3.105)$$

Since particles are diamagnetic, $\dot{\theta} = -\omega_c$. Because of the azimuthal symmetry, P_θ will be a constant of the motion and so

$$const. = P_\theta = -mr^2\omega_c + qr^2B/2 = -\frac{mv_\theta^2}{2\omega_c} = -\frac{m}{q}\mu. \quad (3.106)$$

This shows that constancy of canonical angular momentum is equivalent to μ conservation. It is important to realize that constancy of angular momentum due to perfect axisymmetry is a much more restrictive assumption than the slowness assumption used for adiabatic invariance.

4. *Adiabatic gas law* – The pressure associated with gyrating particles has dimensionality $N = 2$, i.e., $P = (m/2) \int \mathbf{v}' \cdot \mathbf{v}' f d^2v$, where $\mathbf{v}' = v_x\hat{x} + v_y\hat{y}$ and the $x - y$ plane is the plane of the gyration. Also, the density for a two-dimensional system has units of particles/area, i.e., $n \sim 1/A$. Hence, the pressure will scale as $P \sim v_{T\perp}^2/A$. Since $\gamma = (N + 2)/N = 2$, the adiabatic law, Eq. (2.38), gives

$$const. \sim \frac{P}{n^2} \sim \frac{v_{T\perp}^2}{A} A^2; \quad (3.107)$$

but from the flux conservation property of orbits, $A \sim 1/B$ so Eq. (3.107) becomes

$$\frac{P}{n^2} \sim \frac{v_{T\perp}^2}{B}, \quad (3.108)$$

which is again proportional to μ , since $v_{T\perp}^2$ is proportional to the mean perpendicular thermal energy, i.e., the average of the gyrational energies of the individual particles making up the fluid.

3.5.5 Drift equations, frozen-in flux, and frozen-in field lines

The $\mathbf{E} \times \mathbf{B}$ drift, Eq. (3.57), describes motion of bulk plasma across a magnetic field and so shows that it is possible for plasma to move across magnetic field lines. This means that field lines are not invariably frozen into the plasma as is sometimes stated. Motion of plasma across the magnetic field can occur if no constraint exists preventing establishment of an electric field $\mathbf{E} = -\mathbf{U} \times \mathbf{B}$. The plasma can move in the direction perpendicular to the magnetic field provided the motion is such as to preserve the magnetic flux linked by the plasma.

If the magnetic field is static, then the electric field is constrained to be a potential electric field. As an example of a plasma moving across a static magnetic field consider a cylindrical plasma immersed in a static magnetic field $\mathbf{B} = B\hat{z}$ and a radial static electric field $\mathbf{E} = E_r\hat{r}$. The plasma will rotate with a velocity $\mathbf{U} = -\hat{\theta}E_r/B$ and so will be moving across the static magnetic field lines. This motion violates the statement that plasma is frozen to magnetic field lines, but is consistent with the statement that magnetic flux is frozen into the frame of the plasma. This azimuthal rotation of a plasma across a static magnetic field is quite common in laboratory plasmas and occurs spontaneously because of the development of an ambipolar radial electric field established by the difference in the outward radial diffusion rates of electrons and ions. Symmetry considerations or boundary conditions can also constrain the electric field. For example, if the configuration is azimuthally symmetric so $\partial/\partial\theta = 0$, then no azimuthal electrostatic electric field $E_\theta = -r^{-1}\partial\phi/\partial\theta$ is allowed, in which case an ideal plasma cannot move in the radial direction across a constant magnetic field $\mathbf{B} = B\hat{z}$.

If the magnetic field is time-dependent, then an inductive electric field will exist. The magnetic field lines can now be construed as moving, and the drift motion due to the inductive electric field will be such as to keep the plasma attached to the moving field lines as illustrated in Fig. 3.6. This motion will also be such as to preserve the flux linked by the plasma. Thus, a reasonable way of stating the frozen-in field concept is to say that plasma is frozen to moving magnetic field lines, but the converse is not necessarily true, i.e., field lines are not necessarily frozen to a moving plasma. These latter situations involve establishment of a static electric field, i.e., $\mathbf{E} = -\mathbf{U} \times \mathbf{B}$, where $\nabla \times \mathbf{E} = 0$, in which case the plasma moves across the magnetic field but the magnetic field does not change and so the magnetic field lines cannot be construed as moving. If the motion of the plasma across the magnetic field is such that $\nabla \times (\mathbf{U} \times \mathbf{B})$ is not zero, then there will have to be a finite $\partial\mathbf{B}/\partial t$, and so the field lines will move. The motion of the field lines will be such as to preserve the flux linked by any element of plasma.

3.5.6 Magnetic mirrors - a consequence of μ conservation

Consider a charged particle moving in a static, but spatially non-uniform magnetic field. The non-uniformity is such that the field strength varies in the direction of the field line so $\partial B/\partial s \neq 0$, where s is the distance along a field line. Such a field cannot be straight because if it were and so had the form $\mathbf{B} = B_z(z)\hat{z}$, it would necessarily have a non-zero divergence, i.e., it would have $\nabla \cdot \mathbf{B} = \partial B_z/\partial z \neq 0$. Because magnetic fields must have zero divergence, the magnetic field vector must have another component besides B_z and this other component must be

spatially non-uniform in order to contribute to the divergence. Hence the field must be curved if the field strength varies along the direction of the field.

This curvature is easy to see by sketching field lines, as shown in Fig. 3.7. The density of field lines is proportional to the strength of the magnetic field and so a gradient of field strength along the field means that the field lines squeeze together as the field becomes stronger. Because magnetic field lines have zero divergence they are endless and so must bend as they squeeze together. This means that if $\partial B_z / \partial z \neq 0$ there must also be a field transverse to the initial direction of the magnetic field, i.e., a field in the x or y direction. In a cylindrically symmetric system, this transverse field must be a radial field as indicated by the vector decomposition $\mathbf{B} = B_z \hat{z} + B_r \hat{r}$ in Fig. 3.7.

The magnetic field is assumed to be static so that $\nabla \times \mathbf{E} = 0$, in which case $\mathbf{E} = -\nabla\phi$ and Eq. (3.92) can be written as

$$m \frac{dv_{\parallel}}{dt} = -q \frac{\partial \phi}{\partial s} - \mu \frac{\partial B}{\partial s}. \quad (3.109)$$

Multiplying Eq. (3.109) by $v_{\parallel}$ gives

$$\frac{d}{dt} \left[\frac{mv_{\parallel}^2}{2} + q\phi + \mu B \right] = 0, \quad (3.110)$$

assuming that the electrostatic potential is also constant in time. Time integration gives

$$\frac{mv_{\parallel}^2}{2} + q\phi(s) + \mu B(s) = \text{const}. \quad (3.111)$$

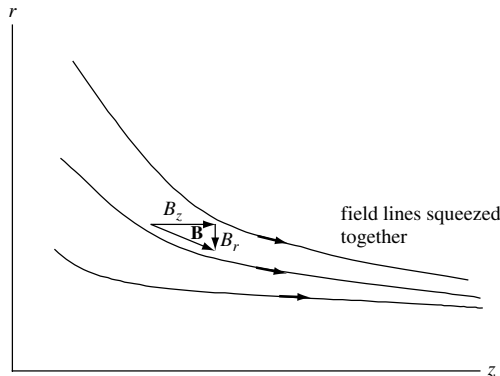


Fig. 3.7 Field lines squeezing together when $\mathbf{B}$ has a gradient. B field is stronger on the right than on the left because density of field lines is larger on the right.

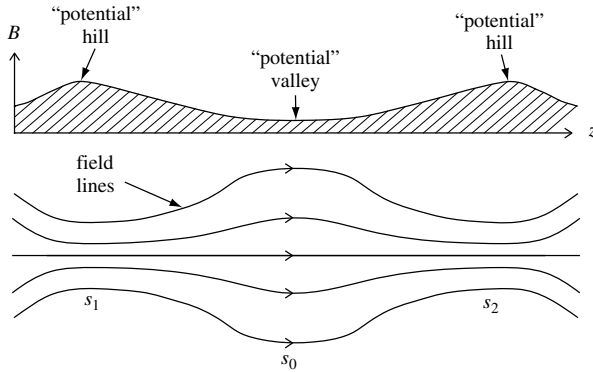


Fig. 3.8 Magnetic mirror.

Thus, $\mu B(s)$ acts as an effective potential energy since it adds to the electrostatic potential energy $q\phi(s)$. This property has the consequence that if $B(s)$ has a minimum with respect to s as shown in Fig. 3.8, then μB acts as an effective potential well that can trap particles. A magnetic trap of this sort can be produced by two axially separated coaxial coils. On each field line $B(s)$ has maxima at the locations s_1 and s_2 near the coils, a minimum at location s_0 between the coils, and $B(s)$ tends to zero as $s \rightarrow \pm\infty$. To focus attention on magnetic trapping, suppose now no electrostatic potential exists so Eq. (3.111) reduces to

$$\frac{mv_{\parallel}^2}{2} + \mu B(s) = \text{const.} \quad (3.112)$$

Now consider a particle with parallel velocity $v_{\parallel 0}$ located at the well minimum s_0 at time $t = 0$. Evaluating Eq. (3.112) at $s = 0$, $t = 0$, and then again when the particle is at some arbitrary position s gives

$$\frac{mv_{\parallel}^2(s)}{2} + \mu B(s) = \frac{mv_{\parallel 0}^2}{2} + \mu B(s_0) = \frac{m(v_{\parallel 0}^2 + v_{\perp 0}^2)}{2} = W_0, \quad (3.113)$$

where W_0 is the particle's total kinetic energy at $t = 0$. Solving Eq. (3.113) for $v_{\parallel}(s)$ gives

$$v_{\parallel}(s) = \pm \sqrt{\frac{2}{m} [W_0 - \mu B(s)]}. \quad (3.114)$$

If $\mu B(s) = W_0$ at some position s , then $v_{\parallel}(s)$ must vanish at this position, in which case the particle must reverse its direction of motion just like a pendulum reversing direction when its velocity goes through zero. This velocity reversal

corresponds to a reflection of the particle and so this configuration is called a magnetic *mirror*. A particle can be trapped between two magnetic mirrors; such a configuration is called a magnetic trap or a magnetic well.

If $W_0 > \mu B_{\max}$, where $B_{\max}$ is the magnitude at $s_{1,2}$ then the parallel velocity does not go to zero at the maximum amplitude of the mirror field. In this case the particle does not reflect, but instead escapes over the peak of the $\mu B(s)$ potential hill and travels out to infinity. Thus, there are two classes of particles:

1. trapped particles – these have $W_0 < \mu B_{\max}$ and bounce back and forth between the mirrors of the magnetic well,
2. untrapped (or passing) particles – these have $W_0 > \mu B_{\max}$ and are retarded at the potential hills but not reflected.

Since $\mu = mv_{\perp 0}^2/2B_{\min}$ and $W_0 = mv_0^2/2$, the criterion for trapping can be written as

$$\frac{B_{\min}}{B_{\max}} < \frac{v_{\perp 0}^2}{v_0^2}. \quad (3.115)$$

Let us define θ as the angle the velocity vector makes with respect to the magnetic field at s_0 , i.e., $\sin \theta = v_{\perp 0}/v_0$, and also define

$$\theta_{\text{trap}} = \sin^{-1} \sqrt{\frac{B_{\min}}{B_{\max}}}. \quad (3.116)$$

Thus, as shown in Fig. 3.9, all particles with $\theta > \theta_{\text{trap}}$ are trapped, while all particles with $\theta < \theta_{\text{trap}}$ are untrapped. Suppose at $t = 0$ the particle velocity distribution at s_0 is isotropic. After a time interval long enough for all untrapped particles to have escaped the trap, there will be no particles in the $\theta < \theta_{\text{trap}}$ region of velocity space. The velocity distribution will thus be zero for $\theta < \theta_{\text{trap}}$; such a distribution function is called a *loss-cone* distribution function.

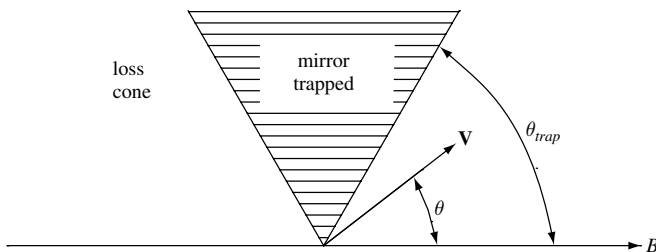


Fig. 3.9 Loss-cone velocity distribution. Particles with velocity angle $\theta > \theta_{\text{trap}}$ are mirror trapped, others are lost.

3.5.7 $\mathcal{J}$, the second adiabatic invariant

Trapped particles have periodic motion in the magnetic well. Thus, applying the concept of adiabatic invariance presented in Section 3.4.1, the quantity

$$\mathcal{J} = \oint P_{\parallel} ds = \oint (mv_{\parallel} + qA_{\parallel}) ds \quad (3.117)$$

will be an invariant if:

1. any time dependence of the well shape is slow compared to the bounce frequency of the trapped particle,
2. any spatial inhomogeneities of the well magnetic field are so gradual that the particle's bounce trajectory changes by only a small amount from one bounce to the next.

To determine the circumstances where $A_{\parallel} \neq 0$, we use Coulomb gauge (i.e., assume $\nabla \cdot \mathbf{A} = 0$) and at any given location define a local Cartesian coordinate system with z axis parallel to the local field. From Ampère's law it is seen that

$$[\nabla \times (\nabla \times \mathbf{A})]_z = -\nabla^2 A_z = \mu_0 J_z \quad (3.118)$$

so A_z is finite only if there is a *current parallel to the magnetic field*. Because J_z acts as the source term in a Poisson-like partial differential equation for A_z , the parallel current need not be at the same location as A_z . If there are no currents parallel to the magnetic field anywhere then $A_{\parallel} = 0$, and in this case the second adiabatic invariant reduces to

$$\mathcal{J} = m \oint v_{\parallel} ds. \quad (3.119)$$

Having a current flow along the magnetic field corresponds to a more complicated magnetic topology. The axial current produces an associated azimuthal magnetic field, which links the axial magnetic field resulting in a helical twist. This more complicated situation of finite magnetic helicity will be discussed in a later chapter.

3.5.8 Consequences of $\mathcal{J}$ -invariance

Just as μ -invariance was related to the perpendicular CGL adiabatic invariant discussed in Section 2.6.5, $\mathcal{J}$ -invariance is closely related to the parallel CGL adiabatic invariant also discussed in Section 2.6.5. To see this, recall that density in a one-dimensional system has dimensions of particles per unit length, i.e., $n_{1D} \sim 1/L$, and pressure in a one-dimensional system has dimensions of kinetic

energy per unit length, i.e., $P_{1D} \sim v_{\parallel}^2/L$. For parallel motion the number of dimensions is $N = 1$ so $\gamma = (N + 2)/N = 3$ and the fluid adiabatic relation is

$$const. \sim \frac{P_{1D}}{n_{1D}^3} \sim \frac{v_{\parallel}^2/L}{L^{-3}} \sim (v_{\parallel}L)^2, \quad (3.120)$$

which is a simplified form of Eq. (3.119) since Eq. (3.119) has the scaling $\mathcal{J} \sim v_{\parallel}L = const.$

$\mathcal{J}$ -invariance combined with mirror trapping/detrapping is the basis of an acceleration mechanism proposed by Fermi (1954) as a means for accelerating cosmic-ray particles to ultra-relativistic velocities. The Fermi mechanism works as follows: consider a particle initially trapped in a magnetic mirror. This particle has an initial angle in velocity space $\theta > \theta_{trap}$; both θ and θ_{trap} are measured when the particle is at the mirror minimum. Now suppose the distance between the magnetic mirrors is slowly reduced so that the bounce distance L of the mirror-trapped particle slowly decreases. This would typically occur by reducing the axial separation between the coils producing the magnetic mirror field. Because $\mathcal{J} \sim v_{\parallel}L$ is invariant, the particle's parallel velocity increases on each successive bounce as L slowly decreases. This steady increase in $v_{\parallel}$ means that the velocity angle θ decreases. Eventually, θ becomes smaller than θ_{trap} whereupon the particle becomes detrapped and escapes from one end of the mirror with a large parallel velocity. This mechanism provides a slow pumping to very high energy, followed by a sudden and automatic ejection of the energetic particle.

3.5.9 The third adiabatic invariant

Consider a particle bouncing back and forth in either of the two geometries shown in Fig. (3.10). In Fig. 3.10(a), the magnetic field is produced by a single

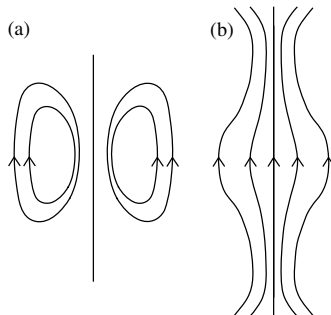


Fig. 3.10 Magnetic field lines relevant to discussion of third adiabatic invariant: (a) field lines always have same curvature (dipole field), (b) field lines have both concave and convex curvature (mirror field).

magnetic dipole and the field lines always have convex curvature, i.e., the radius of curvature is always on the inside of the field lines. The field decreases in magnitude with increasing distance from the dipole.

In Fig. 3.10(b), the field is produced by two coils and has convex curvature near the mirror minimum and concave curvature in the vicinity of the coils. On defining a cylindrical coordinate system (r, θ, z) with z axis coaxial with the coils, it is seen that in the region between the two coils where the field bulges out, the field strength is a decreasing function of r , i.e., $\partial B/\partial r < 0$, whereas in the plane of each coil the opposite is true. Thus, in the mirror minimum, both the centrifugal and grad B forces are radially outward, whereas the opposite is true near the coils.

In both Figs. 3.10(a) and (b) a particle moving along the field line can be mirror-trapped because in both cases the field has a minimum flanked by two maxima. However, for Fig. 3.10(a), the particle will have grad B and curvature drifts always in the same azimuthal sense, whereas for Fig. 3.10(b) the azimuthal direction of these drifts will depend on whether the particle is in a region of concave or convex curvature. Thus, in addition to the mirror bouncing motion, much slower curvature and grad B drifts also exist, directed along the field binormal (i.e., the direction orthogonal to both the field and its radius of curvature). These higher-order drifts may alternate sign during the mirror bouncing. The binormally directed displacement made by a particle during its i th complete period τ of mirror bouncing is

$$\delta \mathbf{r}_j = \int_0^\tau \mathbf{v} dt, \quad (3.121)$$

where τ is the mirror bounce period and $\mathbf{v}$ is the sum of the curvature and grad B drifts experienced in the course of a mirror bounce. This displacement is due to the cumulative effect of the curvature and grad B drifts experienced during one complete period of bouncing between the magnetic mirrors. The average velocity associated with this slow drifting may be defined as

$$\langle \mathbf{v} \rangle = \frac{1}{\tau} \int_0^\tau \mathbf{v} dt. \quad (3.122)$$

Let us calculate the action associated with a sequence of $\delta \mathbf{r}_j$. This action is

$$S = \sum_j [m \langle \mathbf{v} \rangle + q \mathbf{A}]_j \cdot \delta \mathbf{r}_j, \quad (3.123)$$

where the quantity in square brackets is evaluated on the line segment $\delta \mathbf{r}_j$. If the $\delta \mathbf{r}_j$ are small then this can be converted into an action “integral” for the path traced out by the $\delta \mathbf{r}_j$. If the $\delta \mathbf{r}_j$ are sufficiently small to behave as differentials,

then we may write them as $\mathbf{dr}_{\text{bounce}}$ and express the summation as an action integral

$$S = \int [m \langle \mathbf{v} \rangle + q\mathbf{A}] \cdot \mathbf{dr}_{\text{bounce}}, \quad (3.124)$$

where it must be remembered that $\langle \mathbf{v} \rangle$ is the *bounce-averaged* velocity. The quantity $m \langle \mathbf{v} \rangle + q\mathbf{A}$ is just the canonical momentum associated with the effective motion along the sequence of line segments $\delta \mathbf{r}_j$. The vector $\mathbf{r}_{\text{bounce}}$ is a vector pointing from the origin to the particle's location at successive bounces and so is the generalized coordinate associated with the bounce-averaged velocity. If the motion resulting from $\langle \mathbf{v} \rangle$ is periodic, we expect S to be an adiabatic invariant. The first term in Eq. (3.124) will be of the order of $mv_{\text{drift}}2\pi r$, where r is the radius of the trajectory described by the $\delta \mathbf{r}_j$. The second term is just $q\Phi$, where Φ is the magnetic flux enclosed by the trajectory. We compare the ratio of these two terms to obtain

$$\frac{\int m \langle \mathbf{v} \rangle \cdot \mathbf{dr}}{\int q\mathbf{A} \cdot \mathbf{dr}} \sim \frac{mv_{\text{drift}}2\pi r}{qB\pi r^2} \sim \frac{v_{\text{drift}}}{\omega_c r} \sim \frac{r_L^2}{r^2}, \quad (3.125)$$

where we have used $v_{\nabla B} \sim v_c \sim v_{\perp}^2/\omega_c r \sim \omega_c r_L^2/r$. Thus, if the Larmor radius is much smaller than the characteristic scale length of the field, the magnetic flux term dominates the action integral and adiabatic invariance corresponds to the particle *staying on a constant flux surface* as its orbit evolves following the various curvature and grad B drifts. This third adiabatic invariant is much more fragile than $\mathcal{J}$, which in turn was more fragile than μ , because the analysis here is based on the rather strong assumption that the curvature and grad B drifts are small enough for the $\delta \mathbf{r}_j$ to trace out a nearly periodic orbit.

3.6 Relation of drift equations to the double adiabatic MHD equations

The derivation of the MHD Ohm's law involved dropping the Hall term (see p. 55) and the basis for dropping this term was assuming $\omega \ll \omega_{ci}$, where ω is the characteristic rate of change of the electromagnetic field. The derivation of the single particle drift equations involved essentially the same assumption (i.e., the motion was slow compared to ω_{ci}). Thus, if the characteristic rate of change of the electromagnetic field is slow compared to ω_{ci} , both the MHD and the single particle drift equations ought to be equally valid descriptions of the plasma dynamics. If so, there should be some sort of a correspondence principle relating these ostensibly different points of view. Preliminary evidence supporting this hypothesis was the observation that the single particle adiabatic invariants μ and $\mathcal{J}$ were respectively related to the perpendicular and parallel double adiabatic MHD

equations. It thus seems reasonable to expect additional connections between the drift equations and the double adiabatic MHD equations.

In fact, an approximate derivation of the double adiabatic MHD equations can be obtained by summing the currents associated with the various particle drifts – providing one additional effect, *diamagnetic current*, is added to this sum. Diamagnetic current is a peculiar concept because it is a consequence of the macroscopic phenomenon of pressure gradients and so has no meaning in the context of a single particle description.

In order to establish this microscopic–macroscopic relationship we begin by recalling from electromagnetic theory² that a magnetic material with density $\mathbf{M}$ of magnetic dipole moments per unit volume has an associated magnetization current

$$\mathbf{J}_M = \nabla \times \mathbf{M}. \quad (3.126)$$

The magnitude of the magnetic moment of a charged particle in a magnetic field was shown in Section 3.5.4 to be μ . Since a magnetic dipole is represented by a vector pointing in the direction of the magnetic field on the axis of the current loop constituting the dipole, the vector magnetic moment of a charged particle gyrating in a magnetic field will be $\mathbf{m} = -\mu \hat{\mathbf{B}}$. The minus sign corresponds to cyclotron motion being diamagnetic, i.e., the magnetic field resulting from cyclotron rotation opposes the original field in which the particle is rotating. For example, an individual ion placed in a magnetic field $\mathbf{B} = B \hat{\mathbf{z}}$ rotates in the negative θ direction, and so the current associated with the ion motion creates a magnetic field pointing in the $-\hat{\mathbf{z}}$ direction inside the ion orbit.

Suppose there exists a large number or ensemble of particles with density n_σ and mean magnetic moment $\bar{\mu}_\sigma$. The density of magnetic moments, or magnetization density, of this ensemble is

$$\mathbf{M} = - \sum_\sigma n_\sigma \bar{\mu}_\sigma \hat{\mathbf{B}} = - \sum_\sigma n_\sigma \left\langle \frac{m_\sigma v_\perp^2}{2B} \right\rangle \hat{\mathbf{B}} = - \frac{P_\perp \hat{\mathbf{B}}}{B}, \quad (3.127)$$

where $\langle \rangle$ denotes averaging over the velocity distribution and Eq. (2.26) has been used. Inserting Eq. (3.127) into Eq. (3.126) shows that this ensemble of charged particles in a magnetic field has a *diamagnetic current*

$$\mathbf{J}_M = -\nabla \times \left(\frac{P_\perp \hat{\mathbf{B}}}{B} \right). \quad (3.128)$$

² For example, see p. 192 of (Jackson 1998).

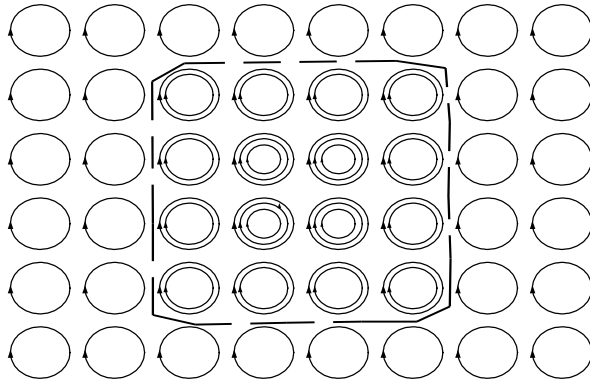


Fig. 3.11 Gradient of magnetized particles gives apparent current as observed on dashed line.

Figure 3.11 shows the physical origin of $\mathbf{J}_M$. Here, a collection of ions all rotate clockwise in a magnetic field pointing out of the page. The azimuthally directed current on the dashed curve is the sum of contributions from (i) particles with guiding centers located one Larmor radius *inside* the dashed curve and (ii) particles with guiding centers located one Larmor radius *outside* the dashed curve. From the point of view of an observer located on the dashed curve, the inside particles (group (i)) constitute a clockwise current, whereas the outside (group (ii)) particles constitute a counterclockwise current. If there are unequal numbers of inside and outside particles (indicated here by concentric circles inside the dashed curve), then the two opposing currents do not cancel and a net macroscopic current appears to flow around the dashed curve, even though *no* actual particles flow around the dashed curve. Inequality of the numbers of inside and outside particles corresponds to a density gradient and so we see that a radial density gradient of gyrating particles gives a net macroscopic azimuthal current. Similarly, if there is a radial temperature gradient, the velocities of the inner and outer groups differ, resulting again in an apparent macroscopic azimuthal current. The combination of density and temperature gradients is such that the net macroscopic current depends on the pressure gradient as given by Eq. (3.128).

Taking diamagnetic current into account is critical for establishing a correspondence between the single particle drifts and the MHD equations, and having recognized this, we are now in a position to derive this correspondence. In order for the derivation to be tractable yet non-trivial, it will be assumed that the magnetic field is time-independent, but the electric field will be allowed to depend on time. It is also assumed that the *dominant* cross-field particle motion is the $\mathbf{v}_E = \mathbf{E} \times \mathbf{B}/B^2$ drift; this assumption is consistent with the hierarchy of particle drifts (i.e., polarization drift is a higher-order correction to $\mathbf{v}_E$).

Because both species have the same $\mathbf{v}_E$, no macroscopic current results from $\mathbf{v}_E$, and so all cross-field currents must result from the other, smaller drifts, namely $\mathbf{v}_{\nabla B}$, $\mathbf{v}_c$, and $\mathbf{v}_p$. Let us now add the magnetization current to the currents associated with these other drifts to obtain the total macroscopic current

$$\mathbf{J}_{\text{total}} = \mathbf{J}_M + \mathbf{J}_{\nabla B} + \mathbf{J}_c + \mathbf{J}_p = \mathbf{J}_M + \sum_{\sigma} n_{\sigma} q_{\sigma} (\mathbf{u}_{\nabla B, \sigma} + \mathbf{u}_{c, \sigma} + \mathbf{u}_{p, \sigma}), \quad (3.129)$$

where $\mathbf{J}_{\nabla B}$, $\mathbf{J}_c$, $\mathbf{J}_p$ are currents due to $\text{grad } B$, curvature, and polarization drifts respectively and $\mathbf{u}_{\nabla B, \sigma}$, $\mathbf{u}_{c, \sigma}$, and $\mathbf{u}_{p, \sigma}$ are the mean (i.e., fluid) velocities associated with these drifts. These currents are explicitly:

1. $\text{grad } B$ current

$$\begin{aligned} \mathbf{J}_{\nabla B} &= \sum_{\sigma} n_{\sigma} q_{\sigma} \mathbf{u}_{\nabla B, \sigma} \\ &= - \sum_{\sigma} \frac{m_{\sigma} n_{\sigma} q_{\sigma} \langle v_{\perp \sigma}^2 \rangle}{2B} \frac{\nabla B \times \mathbf{B}}{q_{\sigma} B^2} = -P_{\perp} \frac{\nabla B \times \mathbf{B}}{B^3}; \end{aligned} \quad (3.130)$$

2. curvature current

$$\begin{aligned} \mathbf{J}_c &= \sum_{\sigma} n_{\sigma} q_{\sigma} \mathbf{u}_{c, \sigma} \\ &= - \sum_{\sigma} n_{\sigma} q_{\sigma} m_{\sigma} \langle v_{\parallel \sigma}^2 \rangle \frac{\hat{B} \cdot \nabla \hat{B} \times \mathbf{B}}{q_{\sigma} B^2} = -P_{\parallel} \frac{\hat{B} \cdot \nabla \hat{B} \times \mathbf{B}}{B^2}; \end{aligned} \quad (3.131)$$

3. polarization current

$$\mathbf{J}_p = \sum_{\sigma} n_{\sigma} q_{\sigma} \mathbf{u}_{p, \sigma} = \sum_{\sigma} n_{\sigma} q_{\sigma} \left(\frac{m_{\sigma}}{q_{\sigma} B^2} \frac{d\mathbf{E}_{\perp}}{dt} \right) = \frac{\rho}{B^2} \frac{d\mathbf{E}_{\perp}}{dt}. \quad (3.132)$$

Because the magnetic field was assumed to be constant, the time derivative of $\mathbf{v}_E$ is the only contributor to the polarization drift current.

The total magnetic force is

$$\begin{aligned} \mathbf{J}_{\text{total}} \times \mathbf{B} &= (\mathbf{J}_M + \mathbf{J}_{\nabla B} + \mathbf{J}_c + \mathbf{J}_p) \times \mathbf{B} \\ &= \left[-\nabla \times \left(\frac{P_{\perp} \hat{B}}{B} \right) - P_{\perp} \frac{\nabla B \times \mathbf{B}}{B^3} \right. \\ &\quad \left. - P_{\parallel} \frac{\hat{B} \cdot \nabla \hat{B} \times \mathbf{B}}{B^2} + \frac{\rho}{B^2} \frac{d\mathbf{E}}{dt} \right] \times \mathbf{B}. \end{aligned} \quad (3.133)$$

The grad B current cancels *part* of the magnetization current as follows:

$$\begin{aligned}\nabla \times \left(\frac{P_{\perp} \hat{B}}{B} \right) + P_{\perp} \frac{\nabla B \times \mathbf{B}}{B^3} &= \left[\nabla \left(\frac{1}{B} \right) \times P_{\perp} \hat{B} + \frac{1}{B} \nabla \times (P_{\perp} \hat{B}) \right] \\ &\quad + P_{\perp} \frac{\nabla B \times \mathbf{B}}{B^3} \\ &= \frac{1}{B} \nabla \times (P_{\perp} \hat{B}) = \frac{P_{\perp}}{B} \nabla \times \hat{B} + \frac{\nabla P_{\perp} \times \hat{B}}{B}\end{aligned}\quad (3.134)$$

so

$$\mathbf{J}_{total} \times \mathbf{B} = - \left[P_{\perp} \nabla \times \hat{B} + \nabla P_{\perp} \times \hat{B} + P_{\parallel} \hat{B} \cdot \nabla \hat{B} \times \hat{B} - \frac{\rho}{B} \frac{d\mathbf{E}}{dt} \right] \times \hat{B}. \quad (3.135)$$

The first term on the right-hand side of Eq. (3.135) can be recast using the vector identity

$$\nabla \left(\frac{\hat{B} \cdot \hat{B}}{2} \right) = 0 = \hat{B} \cdot \nabla \hat{B} + \hat{B} \times \nabla \times \hat{B} \quad (3.136)$$

and the electric field in the last term of Eq. (3.135) can be replaced using $\mathbf{E} = -\mathbf{U} \times \mathbf{B}$ to give

$$\mathbf{J}_{total} \times \mathbf{B} = - (P_{\perp} - P_{\parallel}) \hat{B} \cdot \nabla \hat{B} + \nabla_{\perp} P_{\perp} - \frac{\rho}{B} \frac{d(\mathbf{U} \times \mathbf{B})}{dt} \times \hat{B}. \quad (3.137)$$

The relation $\left[\hat{B} \cdot \nabla \hat{B} \right]_{\perp} = \hat{B} \cdot \nabla \hat{B}$ has been used here; this relation follows from Eq. (3.136). Finally, it is observed that

$$\left[\nabla \cdot (\hat{B} \hat{B}) \right]_{\perp} = \left[(\nabla \cdot \hat{B}) \hat{B} + \hat{B} \cdot \nabla \hat{B} \right]_{\perp} = \hat{B} \cdot \nabla \hat{B} \quad (3.138)$$

so

$$(P_{\perp} - P_{\parallel}) \hat{B} \cdot \nabla \hat{B} = (P_{\perp} - P_{\parallel}) \left[\nabla \cdot (\hat{B} \hat{B}) \right]_{\perp} = \left[\nabla \cdot \{ (P_{\perp} - P_{\parallel}) \hat{B} \hat{B} \} \right]_{\perp}. \quad (3.139)$$

Furthermore,

$$\frac{\rho}{B} \frac{d(\mathbf{U} \times \mathbf{B})}{dt} \times \hat{B} = - \left[\rho \frac{d\mathbf{U}}{dt} \right]_{\perp}, \quad (3.140)$$

since it has been assumed the magnetic field is time-independent. Inserting these last two results in Eq. (3.137) gives

$$\mathbf{J}_{total} \times \mathbf{B} = \left[\nabla \cdot \{ (P_{\parallel} - P_{\perp}) \hat{B} \hat{B} \} \right]_{\perp} + \nabla_{\perp} P_{\perp} + \left[\rho \frac{d\mathbf{U}}{dt} \right]_{\perp} \quad (3.141)$$

or

$$\left[\rho \frac{d\mathbf{U}}{dt} \right]_{\perp} = \left[\mathbf{J}_{total} \times \mathbf{B} - \nabla \cdot \{ P_{\perp} \overleftrightarrow{\mathbf{I}} + (P_{\parallel} - P_{\perp}) \hat{B} \hat{B} \} \right]_{\perp}, \quad (3.142)$$

which is just the perpendicular component of the double adiabatic MHD equation of motion. This demonstrates that if diamagnetic current is taken into account, the drift equations for phenomena with characteristic frequencies ω much smaller than ω_{ci} and the double adiabatic MHD equations are equivalent descriptions of plasma dynamics. This analysis also shows one has to be extremely careful when invoking single particle concepts to explain macroscopic behavior, because if diamagnetic effects are omitted, erroneous conclusions can result.

The justification for the designation “polarization current” results from comparing this current to the current flowing through a parallel plate capacitor with dielectric ε . The capacitance of the parallel plate capacitor is $C = \varepsilon A/d$, where A is the cross-sectional area of the capacitor plates and d is the gap between the plates. The charge on the capacitor is $Q = CV$, where V is the voltage across the capacitor plates. The current through the capacitor is $I = dQ/dt$ so

$$I = C \frac{dV}{dt} = \frac{\varepsilon A}{d} \frac{dV}{dt}. \quad (3.143)$$

Because the electric field between the plates is $E = V/d$ and the current density is $J = I/A$, Eq. (3.143) can be expressed as

$$J = \varepsilon \frac{dE}{dt}, \quad (3.144)$$

which gives the alternating current density in a medium with dielectric ε . If this is compared to the polarization current

$$\mathbf{J}_p = \frac{\rho}{B^2} \frac{d\mathbf{E}_\perp}{dt} \quad (3.145)$$

it is seen that the plasma acts like a dielectric medium in the direction perpendicular to the magnetic field and has an effective dielectric constant given by ρ/B^2 .

The polarization current is intimately related to the acceleration of macroscopic volumes of plasma and also to the details of how an $\mathbf{E} \times \mathbf{B}$ drift becomes established. This is shown schematically in Fig. 3.12(a), where a cube of perfectly conducting plasma is placed at rest between two capacitor plates located in a magnetic field $\mathbf{B} = B\hat{z}$. When the capacitor plates are connected to a source of EMF as in Fig. 3.12(b) so that the upper capacitor plate becomes negatively charged and the lower capacitor plate becomes positively charged, a vertical electric field E_y is established. This causes the electrons and the ions to have an $\mathbf{E} \times \mathbf{B}$ drift to the right as sketched in Fig. 3.12(b) and so the bulk plasma moves to the right with the velocity $\mathbf{U} = \mathbf{E} \times \mathbf{B}/B^2 = \hat{x}E_y/B$. The slight upward motion of the ions and the slight downward motion of the electrons polarizes the cube so a positive surface charge develops on the upper surface of the cube and a negative surface charge develops on the lower surface. The surface polarization creates an electric field opposing the vacuum field due to the capacitor plates and so

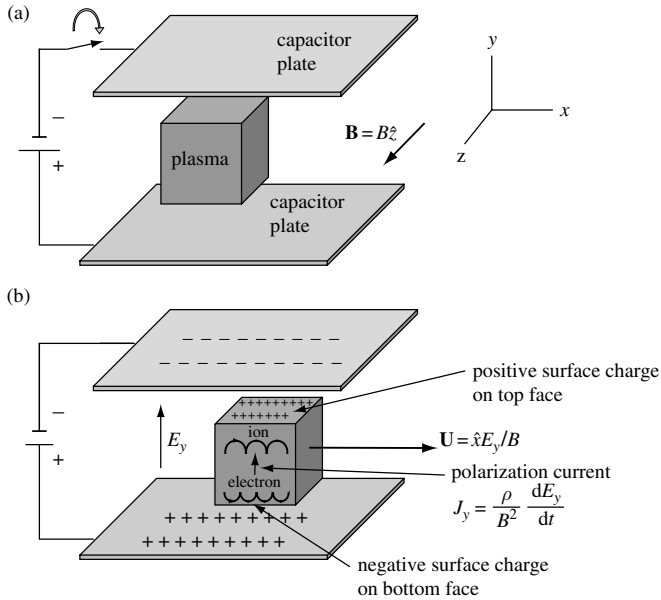


Fig. 3.12 (a) A plasma cube is initially at rest between two capacitor plates located in a magnetic field $\mathbf{B} = B_z \hat{z}$. A source of EMF is connected to the capacitor plates so as to establish a vertical electric field E_y . (b) The $\mathbf{E} \times \mathbf{B}$ motion of ions causes a positive surface charge to develop on the top of the cube while the corresponding electron $\mathbf{E} \times \mathbf{B}$ motion creates a negative surface charge on the bottom of the cube. This surface charge is a polarization charge and reduces the strength of the applied electric field. The polarization current establishes the polarization charge.

the net electric field is reduced relative to the vacuum level, just as the electric field inside a capacitor is reduced by the polarization of the dielectric in the capacitor. The polarization current is the means by which these surface charges are established. The polarization current $\mathbf{J}_p$ flows transiently in the y direction as the cube is accelerated from rest to its steady-state velocity, and this acceleration corresponds to the temporal increase of E_y . This entire process can be equivalently characterized using the MHD point of view, in which case the acceleration of the cube to the right is interpreted as resulting from the cross-product between the polarization current $\mathbf{J}_p$ and the magnetic field $\mathbf{B}$, since

$$\rho \frac{d\mathbf{U}}{dt} = \mathbf{J}_p \times \mathbf{B} = \frac{\rho}{B^2} \frac{d\mathbf{E}_\perp}{dt} \times \mathbf{B} = \rho \frac{d}{dt} \left(\frac{\mathbf{E} \times \mathbf{B}}{B^2} \right). \quad (3.146)$$

The importance of symmetry becomes apparent if one changes the geometrical arrangement to cylindrical geometry $\{r, \phi, z\}$ with $\mathbf{B} = B_z \hat{z}$ and attempts to have radial acceleration of an annular shell of perfectly conducting plasma, i.e., a plasma

shell defined by $a < r < b$, where a is the inner radius and b is the outer radius. The annular shell corresponds to letting $x \rightarrow r$ and $y \rightarrow \phi$ when relating to the situation sketched in Fig. 3.12. Let us imagine the annular shell is decomposed into N equal angular sectors, so that the first sector lies between $\phi = 0$ and $\phi = 2\pi/N$, etc. Viewed individually, each annulus sector should behave in a manner similar to the cube sketched in Fig. 3.12 and develop a surface polarization charge on its ϕ faces, corresponding to the development of an azimuthal electric field E_ϕ . For example, radial motion of the first sector would cause negative polarity to develop on the $\phi = 0$ surface and positive polarity to develop on the $\phi = 2\pi/N$ surface. However, the next sector (i.e., the sector lying between $\phi = 2\pi/N$ and $\phi = 4\pi/N$) would have a negative surface charge at $\phi = 2\pi/N$, which would *cancel* the positive surface charge on the first sector. The azimuthal symmetry essentially short-circuits the development of a finite electrostatic E_ϕ and so prevents the plasma from moving in the radial direction. Thus, whether or not a plasma can move across a magnetic field is intimately related to the symmetry of the configuration. Moving across a magnetic field requires establishment of a polarization surface charge and associated electrostatic field and this cannot occur if prohibited by considerations of symmetry.

3.7 Non-adiabatic motion in symmetric geometry

Adiabatic behavior occurs when temporal or spatial changes in the electromagnetic field from one cyclical orbit to the next are sufficiently gradual to be effectively continuous and differentiable (i.e., analytic). Thus, adiabatic behavior corresponds to situations where variations of the electromagnetic field are sufficiently gradual to be characterized by the techniques of calculus (differentials, limits, Taylor expansions, etc.).

Non-adiabatic particle motion occurs when this is not so. It is therefore no surprise that it is usually not possible to construct analytic descriptions of non-adiabatic particle motion. However, there exist certain special situations where non-adiabatic motion can be described analytically. Using these special cases as a guide, it is possible to develop an understanding for what happens when motion is non-adiabatic.

One special situation is when the electromagnetic field is *geometrically symmetric* with respect to some coordinate Q_j , in which case the symmetry makes it possible to develop analytic descriptions of non-adiabatic motion. This is because symmetry in Q_j causes the canonical momentum P_j to be an exact constant of the motion. The critical feature is that P_j remains constant no matter how drastically the field changes in time or space because Lagrange's equation $\dot{P}_j = -\partial L / \partial Q_j$ has no limitations on the rate at which changes can occur. In effect, being symmetric

trumps being non-analytic. The absolute invariance of P_j when $\partial L / \partial Q_j = 0$ reduces the number of equations and allows a partial or sometimes even a complete solution of the motion. Solutions to symmetric problems give valuable insight regarding the more general situation of being both non-adiabatic and asymmetric.

Two closely related examples of non-adiabatic particle motion will now be analyzed: (i) sudden temporal and (ii) sudden spatial reversal of the polarity of an azimuthally symmetric magnetic field having no azimuthal component. The most general form of such a field can be written in cylindrical coordinates (r, θ, z) as

$$\mathbf{B} = \frac{1}{2\pi} \nabla \psi(r, z, t) \times \nabla \theta; \quad (3.147)$$

a field of this form is called poloidal. Rather than using $\hat{\theta}$ explicitly, the form $\nabla \theta$ has been used because $\nabla \theta$ is better suited for use with the various identities of vector calculus (e.g., $\nabla \times \nabla \theta = 0$) and leads to greater algebraic clarity. The relationship between $\nabla \theta$ and $\hat{\theta}$ is seen by simply taking the gradient:

$$\nabla \theta = \left(\hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \hat{z} \frac{\partial}{\partial z} \right) \theta = \frac{\hat{\theta}}{r}. \quad (3.148)$$

Equation (3.147) automatically satisfies $\nabla \cdot \mathbf{B} = 0$ (by virtue of the vector identity $\nabla \cdot (\mathbf{G} \times \mathbf{H}) = \mathbf{H} \cdot \nabla \times \mathbf{G} - \mathbf{G} \cdot \nabla \times \mathbf{H}$), has no θ component, and is otherwise arbitrary, since ψ is arbitrary. As shown in Fig. 3.13, the magnetic flux linking a circle of radius r with center at axial position z is

$$\begin{aligned} \int \mathbf{B} \cdot d\mathbf{s} &= \int_0^r 2\pi r dr \hat{z} \cdot \left[\frac{1}{2\pi} \nabla \psi(r, z, t) \times \nabla \theta \right] \\ &= \int_0^r dr \frac{\partial \psi(r, z, t)}{\partial r} = \psi(r, z, t) - \psi(0, z, t). \end{aligned} \quad (3.149)$$

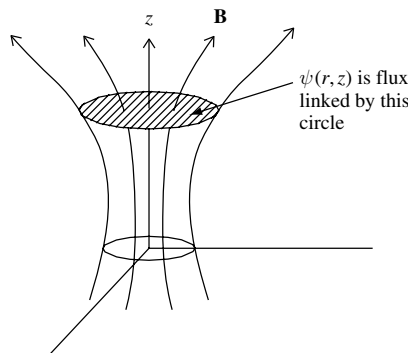


Fig. 3.13 Azimuthally symmetric flux surface.

However,

$$B_r(r, z, t) = -\frac{1}{2\pi r} \frac{\partial \psi}{\partial z} \quad (3.150)$$

and since $\nabla \cdot \mathbf{B} = 0$, B_r must vanish at $r = 0$, and so $\partial \psi / \partial z = 0$ on the symmetry axis $r = 0$.

Thus ψ is constant along the symmetry axis $r = 0$; for convenience we choose this constant to be zero. Hence, $\psi(r, z, t)$ is precisely the magnetic flux enclosed by a circle of radius r at axial location z . We can also use the vector potential $\mathbf{A}$ to calculate the magnetic flux through the same circle and obtain

$$\int \mathbf{B} \cdot d\mathbf{s} = \int \nabla \times \mathbf{A} \cdot d\mathbf{s} = \oint \mathbf{A} \cdot d\mathbf{l} = \int_0^{2\pi} A_\theta r d\theta = 2\pi r A_\theta. \quad (3.151)$$

This shows that the flux ψ and the vector potential A_θ are related by

$$\psi(r, z, t) = 2\pi r A_\theta. \quad (3.152)$$

No other component of vector potential is required to determine the magnetic field and so we may set $\mathbf{A} = A_\theta(r, z, t)\hat{\theta}$.

The current $\mathbf{J} = \mu_0^{-1} \nabla \times \mathbf{B}$ producing this magnetic field is purely azimuthal as can be seen by considering the r and z components of $\nabla \times \mathbf{B}$. The actual current density is

$$\begin{aligned} J_\theta &= \mu_0^{-1} r \nabla \theta \cdot \nabla \times \mathbf{B} \\ &= \mu_0^{-1} r \nabla \cdot (\mathbf{B} \times \nabla \theta) \\ &= -\frac{r}{2\pi\mu_0} \nabla \cdot \left(\frac{1}{r^2} \nabla \psi \right) \\ &= -\frac{r}{2\pi\mu_0} \left[\frac{\partial}{\partial r} \left(\frac{1}{r^2} \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial z^2} \right], \end{aligned} \quad (3.153)$$

a Poisson-like equation. Since no current loops can exist at infinity, the field prescribed by Eq. (3.147) must be produced by a set of coaxial coils having various finite radii r and various finite axial positions z .

The axial magnetic field is

$$B_z = \frac{1}{2\pi r} \frac{\partial \psi}{\partial r}. \quad (3.154)$$

Near $r = 0$, ψ can always be Taylor expanded as

$$\psi(r, z) = 0 + r \frac{\partial \psi(r=0, z)}{\partial r} + \frac{r^2}{2} \frac{\partial^2 \psi(r=0, z)}{\partial r^2} + \dots \quad (3.155)$$

Suppose $\partial \psi / \partial r$ is non-zero at $r = 0$, i.e., $\psi \sim r$ near $r = 0$. If this were the case, then the first term in the right-hand side of the last line of Eq. (3.153) would

become infinite and so lead to an infinite current density at $r = 0$. Such a result is non-physical and so we require that the first non-zero term in the Taylor expansion of ψ about $r = 0$ to be the r^2 term.

Every field line looping through the inside of a current loop also loops back in the reverse direction on the outside, so there is no net magnetic flux at infinity. This means ψ must vanish at infinity and so as r increases, ψ increases from its value of zero at $r = 0$ to some maximum value $\psi_{\max}$ at $r = r_{\max}$, and then slowly decreases back to zero as $r \rightarrow \infty$. As seen from Eq. (3.154) this corresponds to B_z being positive for $r < r_{\max}$ and negative for $r > r_{\max}$. A contour plot of the $\psi(r, z)$ flux surfaces and a plot of $\psi(r, z = 0)$ versus r is shown in Fig. 3.14.

In this cylindrical coordinate system the Lagrangian, Eq. (3.12), has the form

$$L = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2) + qr\dot{\theta}A_\theta - q\phi(r, z, t). \quad (3.156)$$

Since θ is an ignorable coordinate, the canonical angular momentum is a constant of the motion, i.e.,

$$P_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2 \dot{\theta} + qrA_\theta = \text{const.} \quad (3.157)$$

or, in terms of flux,

$$P_\theta = mr^2 \dot{\theta} + \frac{q}{2\pi} \psi(r, z, t) = \text{const.} \quad (3.158)$$

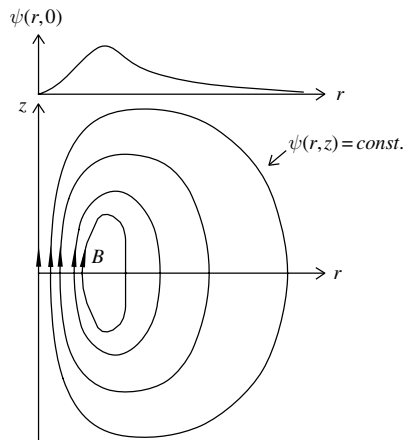


Fig. 3.14 Contour plot of flux surfaces.

Thus, the Hamiltonian is

$$\begin{aligned}
 H &= \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2) + \phi(r, z, t) \\
 &= \frac{m}{2} (\dot{r}^2 + \dot{z}^2) + \frac{(P_\theta - q\psi(r, z, t)/2\pi)^2}{2mr^2} + \phi(r, z, t) \\
 &= \frac{m}{2} (\dot{r}^2 + \dot{z}^2) + \chi(r, z, t),
 \end{aligned} \tag{3.159}$$

where

$$\chi(r, z, t) = \frac{1}{2m} \left[\frac{P_\theta - q\psi(r, z, t)/2\pi}{r} \right]^2 \tag{3.160}$$

is an *effective* potential. For purposes of plotting, the effective potential can be written in a dimensionless form as

$$\frac{\chi(r, z, t)}{\chi_0} = \left(\frac{\frac{2\pi P_\theta}{q\psi_0} - \frac{\psi(r, z, t)}{\psi_0}}{r/L} \right)^2, \tag{3.161}$$

where L is some reference scale length, ψ_0 is some arbitrary reference value for the flux, and $\chi_0 = q\psi_0^2/8\pi^2 L^2 m$. For simplicity we have set $\phi(r, z, t) = 0$, since this term gives the motion of a particle in a readily understood, two-dimensional electrostatic potential.

Suppose that for times $t < t_1$ the coil currents are constant, in which case the associated magnetic field and flux are also constant. Since the Lagrangian does not explicitly depend on time, the energy H is a constant of the motion. Hence, there are two constants of the motion, H and P_θ . Consider now a particle located initially on the midplane $z = 0$ with $r < r_{\max}$. The particle motion depends on the sign of $q\psi/P_\theta$ and so we consider each polarity separately.

1. $q\psi/P_\theta$ is positive. If $2\pi|P_\theta| < |q\psi_{\max}|$ there exists a location inside $r_{\max}$ where

$$P_\theta = \frac{q}{2\pi} \psi \tag{3.162}$$

and there exists a location outside $r_{\max}$ where this equality holds as well. χ vanishes at these two points, which are also local minima of χ because χ is positive-definite. The top plot in Fig. 3.15 shows a nominal $\psi(r)/\psi_0$ flux profile and the middle plot shows the corresponding $\chi(r)/\chi_0$; the z and t dependences are suppressed from the arguments for clarity. There exists a maximum of χ between the two minima. We consider a particle initially located in one of the two minima of χ . If $H < \chi_{\max}$ the

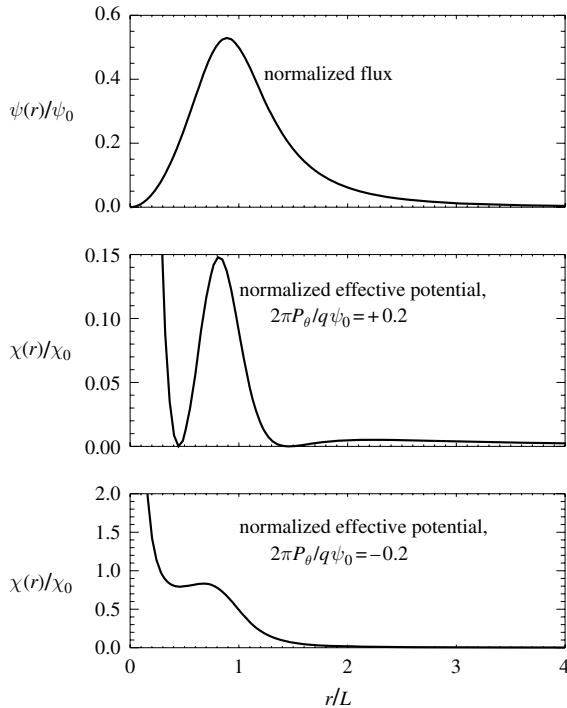


Fig. 3.15 Specific example (with z dependence suppressed) showing ψ and χ relationship: top is plot of function $\psi(r)/\psi_0 = (r/L)^2 / (1 + (r/L)^6)$, middle and bottom plots show corresponding normalized effective potential for $2\pi P_\theta/q\psi_0 = +0.2$ and $2\pi P_\theta/q\psi_0 = -0.2$. Both middle and bottom plots have a minimum at $r/L \simeq 0.45$; middle plot also has a minimum at $r/L \simeq 1.4$. The two minima in the middle plot occur when $\chi(r)/\chi_0 = 0$ but the single minimum in the bottom plot occurs at a finite value of $\chi(r)/\chi_0$, indicating that an axis-encircling particle must have finite energy.

particle will be confined to an effective potential well centered about the flux surface defined by Eq. (3.162). From Eq. (3.158) the angular velocity is

$$\dot{\theta} = \frac{1}{mr^2} \left(P_\theta - \frac{q\psi}{2\pi} \right). \quad (3.163)$$

The sign of $\dot{\theta}$ reverses periodically as the particle bounces back and forth in the χ potential well. This corresponds to localized gyromotion as shown in Fig. 3.16 (left).

2. $q\psi/P_\theta$ is negative. In this case χ can never vanish, because $P_\theta - q\psi/2\pi$ never vanishes. Nevertheless, it is still possible for χ to have a minimum and, hence, a potential well. This possibility can be seen by setting $\partial\chi/\partial r = 0$, which occurs when

$$\left(P_\theta - \frac{q}{2\pi}\psi \right) \frac{\partial}{\partial r} \left(\frac{P_\theta - q\psi/2\pi}{r} \right) = 0. \quad (3.164)$$

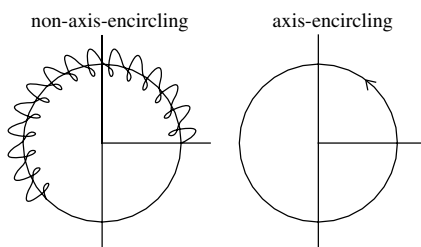


Fig. 3.16 Localized gyromotion associated with particle bouncing in effective potential well.

Equation (3.164) can be satisfied by having

$$\frac{\partial}{\partial r} \left(\frac{P_\theta - \frac{q}{2\pi} \psi}{r} \right) = 0, \quad (3.165)$$

which implies

$$P_\theta = -\frac{qr^2}{2\pi} \frac{\partial}{\partial r} \left(\frac{\psi}{r} \right). \quad (3.166)$$

Recall that ψ had a maximum, $\psi \sim r^2$ near $r = 0$, and also $\psi \rightarrow 0$ as $r \rightarrow \infty$. Thus $\psi/r \sim r$ for small r and $\psi/r \rightarrow 0$ for $r \rightarrow \infty$ so ψ/r also has a maximum; this maximum is located at an r somewhat inside of the maximum of ψ . Equation (3.166) can only be valid at points inside of this maximum; otherwise the assumption of opposite signs for P_θ and ψ would be incorrect. Furthermore, Eq. (3.166) can only be satisfied if $|P_\theta|$ is not too large, because the right-hand side of Eq. (3.166) has a maximum value. If all these conditions are satisfied, then χ will have a non-zero minimum as shown in the bottom plot of Fig. 3.15.

A particularly simple example of this behavior occurs if Eq. (3.166) is satisfied near the $r = 0$ axis (i.e., where $\psi \sim r^2$) so that this equation becomes simply

$$P_\theta = -\frac{q}{2\pi} \psi, \quad (3.167)$$

which is just the opposite of Eq. (3.162). Substituting in Eq. (3.163) we see that $\dot{\theta}$ now *never* changes sign; i.e., the particle is *axis-encircling* as sketched in Fig. 3.16 (right). The Larmor radius of this axis-encircling particle is just the radius of the minimum of the potential well, the radius where Eq. (3.166) holds. The azimuthal kinetic energy of the particle corresponds to the height of the minimum of χ in the bottom plot of Fig. 3.15.

3.7.1 Temporal reversal of magnetic field-energy gain

Armed with this information about axis-encircling and non-axis-encircling particles, we now examine the strongly non-adiabatic situation where a coil current starts at $I = I_0$, is reduced to zero, and then becomes $I = -I_0$, so that all fields and fluxes reverse sign. The particle energy will not stay constant for this situation because the Lagrangian depends explicitly on time. However, since symmetry is maintained, P_θ must remain constant. Thus, a non-axis-encircling particle (with radial location determined by Eq. (3.162)) will change to an axis-encircling particle if a minimum exists for χ when the sign of ψ is reversed. If such a minimum does exist and if the initial radius was near the axis where $\psi \sim r^2$, then comparison of Eqs. (3.162) and (3.167) shows that the particle will have the same radius after the change of sign as before. The particle will gain energy during the field reversal by an amount corresponding to the finite value of the minimum of χ for the axis-encircling case.

This process can also be considered from the point of view of particle drifts: initially, the non-axis-encircling particle is frozen to a constant ψ surface (flux surface). When the coil current starts to decrease, the maximum value of the flux correspondingly decreases. The constant ψ contours on the *inside* of $\psi_{\max}$ move outwards towards the location of $\psi_{\max}$ where they are annihilated. Likewise, the contours outside of $\psi_{\max}$ move inwards to $\psi_{\max}$ where they are also annihilated.

To the extent that the $\mathbf{E} \times \mathbf{B}$ drift is a valid approximation, its effect is to keep the particle attached to a surface of constant flux. This can be seen by integrating Faraday's law over the area of a circle of radius r to obtain $\int ds \cdot \nabla \times \mathbf{E} = -\int ds \cdot \partial \mathbf{B} / \partial t$ and then invoking Stokes' theorem to give

$$E_\theta 2\pi r = -\frac{\partial \psi}{\partial t}. \quad (3.168)$$

The theta component of $\mathbf{E} + \mathbf{v} \times \mathbf{B} = 0$ is

$$E_\theta + v_z B_r - v_r B_z = 0 \quad (3.169)$$

and from (3.147), $B_r = -(2\pi r)^{-1} \partial \psi / \partial z$ and $B_z = -(2\pi r)^{-1} \partial \psi / \partial r$. Combination of Eqs. (3.168) and (3.169) thus gives

$$\frac{\partial \psi}{\partial t} + v_r \frac{\partial \psi}{\partial r} + v_z \frac{\partial \psi}{\partial z} = 0. \quad (3.170)$$

Because $\psi(r(t), z(t), t)$ is the flux measured in the frame of a particle moving with trajectory $r(t)$ and $z(t)$, Eq. (3.170) shows that the $\mathbf{E} \times \mathbf{B}$ drift maintains the particle on a surface of constant ψ , i.e., the $\mathbf{E} \times \mathbf{B}$ drift is such as to

maintain $d\psi/dt = 0$, where d/dt means time derivative as measured in the particle frame.

The implication of this attachment of the particle to a surface of constant ψ can be appreciated by making an analogy to the motion of people initially located on the beach of a volcanic island that is slowly sinking into the sea. In order to avoid being drowned as the island sinks, the people will move towards the mountain top to stay at a constant height above the sea. The location of $\psi_{\max}$ here corresponds to the mountain top and the particles trying to stay on surfaces of constant ψ correspond to people trying to stay at constant altitude. A particle initially located at some location away from the “mountain top” $\psi_{\max}$ moves towards $\psi_{\max}$ if the overall level of all the ψ surfaces is sinking. The reduction of ψ as measured at a fixed position will create the azimuthal electric field given by Eq. (3.168) and this electric field will, as shown by Eqs. (3.169) and (3.170), cause an $\mathbf{E} \times \mathbf{B}$ drift, which convects each particle in just such a way as to stay on a constant ψ contour.

The $\mathbf{E} \times \mathbf{B}$ drift approximation breaks down when $\mathbf{B}$ becomes zero, i.e., when ψ changes polarity. This breakdown corresponds to a breakdown of the adiabatic approximation. If ψ changes polarity before a particle reaches $\psi_{\max}$, the particle becomes axis-encircling. The extra energy associated with being axis-encircling is obtained when $\psi \simeq 0$ but $\partial\psi/\partial t \neq 0$ so that there is an electric field E_θ , but no magnetic field. Finite E_θ and no magnetic field results in a simple theta acceleration of the particle. Thus, when ψ reverses polarity the particle is accelerated azimuthally and develops finite kinetic energy. After ψ has changed polarity the magnitude of ψ increases and the adiabatic approximation again becomes valid. Because the polarity is reversed, increase of the magnitude of ψ is now analogous to creating an ever deepening crater. Particles again try to stay on constant flux surfaces as dictated by Eq. (3.170) and as the crater deepens, the particles have to move away from $\psi_{\min}$ to stay at the same altitude. When the reversed flux attains the same magnitude as the original flux, the flux surfaces have the same shape as before. However, the particles are now axis-encircling and have the extra kinetic energy obtained at field reversal.

3.7.2 Spatial reversal of field-cusps

Suppose two solenoids with constant currents are arranged coaxially with their magnetic fields opposing each other as shown in Fig. 3.17(a). Since the solenoid currents are constant, the Lagrangian does not depend explicitly on time, in which case energy is a constant of the motion. Because of the geometrical arrangement, the flux function is antisymmetric in z , where $z = 0$ defines the midplane between the two solenoids.

Consider a particle injected with initial velocity $\mathbf{v} = v_{z0}\hat{z}$ at $z = -L$, $r = a$. Since this particle has no initial $v_{\perp}$, it simply streams along a magnetic field line. However, when the particle approaches the cusp region, the magnetic field lines start to curve causing the particle to develop both curvature and grad B drifts perpendicular to the magnetic field. When the particle approaches the $z = 0$ plane, the drift approximation breaks down because $B \rightarrow 0$ and so the particle's motion becomes non-adiabatic (cf. Fig. 3.17(a)).

Although the particle trajectory is very complex in the vicinity of the cusp, it is still possible to determine whether the particle will cross into the positive z half-plane, i.e., cross the cusp. Such an analysis is possible because two constants of the motion exist, namely P_{θ} and H . The energy

$$H = \frac{P_r^2}{2m} + \frac{P_z^2}{2m} + \frac{\left(P_{\theta} - \frac{q}{2\pi}\psi(r, z)\right)^2}{2mr^2} = \text{const.} \quad (3.171)$$

can be evaluated using

$$P_{\theta} = \left[mr^2 \dot{\theta} + \frac{q}{2\pi} \psi \right]_{\text{initial}} = \frac{q}{2\pi} \psi_0, \quad (3.172)$$

since initially $\dot{\theta} = 0$. Here

$$\psi_0 = \psi(r = a, z = -L) \quad (3.173)$$

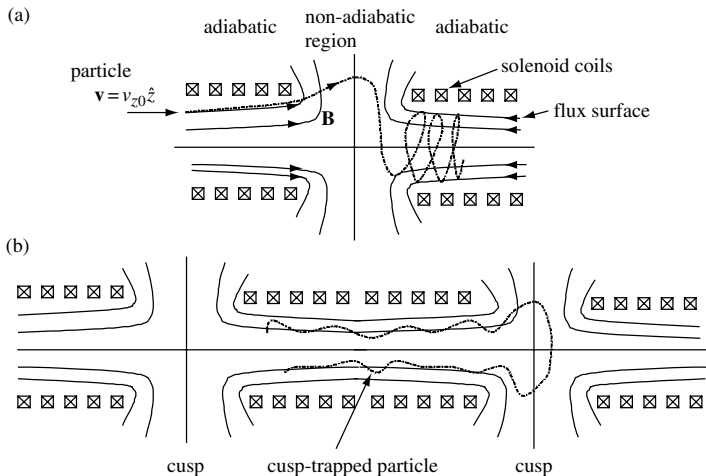


Fig. 3.17 (a) Cusp field showing trajectory for particle with sufficient initial energy to penetrate the cusp; (b) two cusps used as magnetic trap to confine particles.

is the flux at the particle's initial position. Inserting initial values of all quantities in Eq. (3.171) gives

$$H = \frac{mv_{z0}^2}{2} \quad (3.174)$$

and so Eq. (3.171) becomes

$$\begin{aligned} \frac{mv_{z0}^2}{2} &= \frac{mv_r^2}{2} + \frac{mv_z^2}{2} + \frac{\left(\frac{q}{2\pi}\right)^2 (\psi_0 - \psi(r, z))^2}{2mr^2} \\ &= \frac{mv_r^2}{2} + \frac{mv_z^2}{2} + \frac{mv_\theta^2}{2}. \end{aligned} \quad (3.175)$$

The extent to which a particle penetrates the cusp can be easily determined if the particle starts close enough to $r = 0$ so that the flux may be approximated as $\psi \sim r^2$. Specifically, the flux will be $\psi = B_{z0}\pi r^2$, where B_{z0} is the on-axis magnetic field in the $z \ll 0$ region. The canonical momentum is simply $P_\theta = q\psi/2\pi = qB_{z0}a^2/2$, since the particle started as non-axis-encircling.

Suppose the particle penetrates the cusp and arrives at some region where again $\psi \sim r^2$. Since the particle is now axis-encircling, the relation between canonical momentum and flux is $P_\theta = -q\psi/2\pi = -q(-B_{z0}\pi r^2)/2\pi = qB_{z0}r^2/2$ from which it is concluded that $r = a$. Thus, if the particle is able to move across the cusp, it becomes an axis-encircling particle with the *same* radius $r = a$ it originally had when it was non-axis-encircling. The minimum energy an axis-encircling particle can have is when it is purely axis-encircling, i.e., has $v_r = 0$ and $v_z = 0$. Thus, in order for the particle to cross the cusp and reach a location where it becomes purely axis-encircling, the particle's initial energy must satisfy

$$\frac{mv_{z0}^2}{2} \geq \frac{m(\omega_c a)^2}{2} \quad (3.176)$$

or simply

$$v_{z0} \geq \omega_c a. \quad (3.177)$$

If v_{z0} is too small to satisfy this relation, the particle reflects from the cusp and returns back to the negative z half-plane. Plasma confinement schemes have been designed based on particles reflecting from cusps as shown in Fig. 3.17(b). Here a particle is trapped between two cusps and, so long as its parallel energy is insufficient to violate Eq. (3.177), the particle is confined between the two cusps.

Cusps have also been used to trap relativistic electron beams in mirror fields (Kribel *et al.* 1974, Hudgings *et al.* 1978). In this scheme an additional opposing solenoid is added to one end of a magnetic mirror so as to form a cusp outside the mirror region. A relativistic electron beam is injected through the cusp into

the mirror. The beam changes from non-axis-encircling into axis-encircling on passing through the cusp as in Fig. 3.17(a). If energy is conserved, the beam is not trapped because the beam will reverse its trajectory and bounce back out of the mirror. However, if axial energy is removed from the beam once it is in the mirror, then the motion will not be reversible and the beam will be trapped. Removal of beam axial energy has been achieved by having the beam collide with neutral particles or by having the beam induce currents in a resistive wall.

3.7.3 Stochastic motion in large-amplitude, low-frequency waves

The particle drifts ($\mathbf{E} \times \mathbf{B}$, polarization, etc.) were derived using an iteration scheme based on the assumption that spatial changes in the electric and magnetic fields are sufficiently gradual to allow Taylor expansions of the fields about their values at the gyrocenter.

We now examine a situation where the fields change gradually in space relative to the initial gyro-orbit dimensions, but the fields also pump energy into the particle motion so the size of the gyro-orbit eventually increases to the point that the smallness assumption fails. To see how this might occur, consider motion of a particle in an electrostatic wave

$$\mathbf{E} = \hat{y} k \phi \sin(ky - \omega t) \quad (3.178)$$

propagating in a plasma immersed in a uniform magnetic field $\mathbf{B} = B\hat{z}$. The wave frequency is much lower than the cyclotron frequency of the particle in question. This $\omega \ll \omega_c$ condition indicates that, in principle, the drift equations can be used and so according to these equations, the charged particle should have both an $\mathbf{E} \times \mathbf{B}$ drift

$$\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2} = \hat{x} \frac{k\phi}{B} \sin(ky - \omega t) \quad (3.179)$$

and a polarization drift

$$\mathbf{v}_p = \frac{m}{qB^2} \frac{d\mathbf{E}_\perp}{dt} = \hat{y} \frac{mk\phi}{qB^2} \frac{d}{dt} \sin(ky - \omega t). \quad (3.180)$$

If the wave amplitude is infinitesimal, the spatial displacements associated with $\mathbf{v}_E$ and $\mathbf{v}_p$ are negligible and so the guiding center value of y may be used in the right-hand side of Eq. (3.180) to obtain

$$\mathbf{v}_p = -\hat{y} \frac{\omega mk\phi}{qB^2} \cos(ky - \omega t). \quad (3.181)$$

Equations (3.179) and (3.181) show that the combined $\mathbf{v}_E$ and $\mathbf{v}_p$ particle drift motion results in an elliptical trajectory.

Now suppose the wave amplitude becomes so large that the particle is displaced significantly from its initial position. Since the polarization drift is in the y direction, there will be a substantial displacement in the y direction. Thus, the right side of Eq. (3.180) should be construed as $\sin(ky(t) - \omega t)$ so, taking into account the time dependence of y on the right-hand side, Eq. (3.180) becomes

$$\mathbf{v}_p = \hat{y} \frac{mk\phi}{qB^2} \frac{d}{dt} \sin(ky - \omega t) = \hat{y} \frac{mk\phi}{qB^2} \left(k \frac{dy}{dt} - \omega \right) \cos(ky - \omega t). \quad (3.182)$$

However, $dy/dt = v_p$, since $\mathbf{v}_p$ is the motion in the y direction. Equation (3.182) becomes an implicit equation for $\mathbf{v}_p$ and may be solved to give

$$\mathbf{v}_p = -\hat{y} \frac{\omega mk\phi}{qB^2} \frac{\cos(ky - \omega t)}{[1 - \alpha \cos(ky - \omega t)]}, \quad (3.183)$$

where

$$\alpha = \frac{mk^2\phi}{qB^2} \quad (3.184)$$

is a non-dimensional measure of the wave amplitude (McChesney, Stern, and Bellan 1987, White, Chen and Lin 2002).

If $\alpha > 1$, the denominator in Eq. (3.183) vanishes when $ky - \omega t = \cos^{-1} \alpha^{-1}$ and this vanishing denominator would result in an infinite polarization drift. However, the derivation of the polarization drift was based on the assumption that the time derivative of the polarization drift was negligible compared to the time derivative of $\mathbf{v}_E$, i.e., it was explicitly assumed $d\mathbf{v}_p/dt \ll d\mathbf{v}_E/dt$. Clearly, this assumption fails when $\mathbf{v}_p$ becomes infinite and so the iteration scheme used to derive the particle drifts fails. What is happening is that when $\alpha \sim 1$, the particle displacement due to polarization drift becomes $\sim k^{-1}$, i.e., the displacement of the particle from its gyrocenter becomes of the order of a wavelength. In such a situation it is incorrect to represent the particle's actual location by its gyrocenter because the particle experiences the wave field at its actual location, not at its gyrocenter. Because the wave field is significantly different at two locations separated by $\sim k^{-1}$, it is essential to use the wave field evaluated at the actual particle location rather than at the gyrocenter.

Direct numerical integration of the Lorentz equation in this large-amplitude limit shows that when α exceeds unity, particle motion becomes chaotic and cannot be described by analytic formulae. Onset of chaotic motion resembles heating of the particles since chaos and heating both broaden the velocity distribution function. However, chaotic heating is not a true heating because entropy is not increased – the motion is deterministic and not random. Nevertheless, this chaotic (or stochastic) heating is indistinguishable for practical purposes from ordinary collisional thermalization of directed motion.

An alternate way of looking at this issue is to consider the Lorentz equations for two initially adjacent particles, denoted by subscripts 1 and 2, that are in a wave electric field and a uniform, steady-state magnetic field (Stasiewicz, Lundin and Marklund 2000). The respective Lorentz equations of the two particles are

$$\begin{aligned}\frac{d\mathbf{v}_1}{dt} &= \frac{q}{m} [\mathbf{E}(\mathbf{x}_1, t) + \mathbf{v}_1 \times \mathbf{B}] \\ \frac{d\mathbf{v}_2}{dt} &= \frac{q}{m} [\mathbf{E}(\mathbf{x}_2, t) + \mathbf{v}_2 \times \mathbf{B}].\end{aligned}\quad (3.185)$$

Subtracting these two equations gives an equation for the difference between the velocities of the two particles, $\delta\mathbf{v} = \mathbf{v}_1 - \mathbf{v}_2$, in terms of the difference $\delta\mathbf{x} = \mathbf{x}_1 - \mathbf{x}_2$ in their positions, i.e.,

$$\frac{d\delta\mathbf{v}}{dt} = \frac{q}{m} [\delta\mathbf{x} \cdot \nabla \mathbf{E} + \delta\mathbf{v} \times \mathbf{B}]. \quad (3.186)$$

The difference velocity is related to the difference in positions by $d\delta\mathbf{x}/dt = \delta\mathbf{v}$. Let y be the direction in which the electric field is non-uniform, i.e., with this choice of coordinate system $\mathbf{E}$ depends only on the y direction. To simplify the algebra, define $\mathcal{E}_x = qE_x/m$ and $\mathcal{E}_y = qE_y/m$ so the components of Eq. (3.186) transverse to the magnetic field are

$$\begin{aligned}\delta\ddot{x} &= \delta y \frac{\partial \mathcal{E}_x}{\partial y} + \omega_c \delta \dot{y} \\ \delta\ddot{y} &= \delta y \frac{\partial \mathcal{E}_y}{\partial y} - \omega_c \delta \dot{x}.\end{aligned}\quad (3.187)$$

Now take the time derivative of the lower equation to obtain

$$\delta\dddot{y} = \delta \dot{y} \frac{\partial \mathcal{E}_y}{\partial y} + \delta y \frac{\partial}{\partial y} \left(\frac{d\mathcal{E}_y}{dt} \right) - \omega_c \delta \ddot{x} \quad (3.188)$$

and then substitute for $\delta\ddot{x}$ giving

$$\delta\dddot{y} = \delta \dot{y} \frac{\partial \mathcal{E}_y}{\partial y} + \delta y \frac{\partial}{\partial y} \left(\frac{d\mathcal{E}_y}{dt} \right) - \omega_c \left(\delta y \frac{\partial \mathcal{E}_x}{\partial y} + \omega_c \delta \dot{y} \right). \quad (3.189)$$

This can be rearranged as

$$\delta\dddot{y} + \omega_c^2 \left(1 - \frac{1}{\omega_c} \frac{\partial \mathcal{E}_y}{\partial y} \right) \delta \dot{y} = \omega_c \delta y \frac{\partial \mathcal{E}_x}{\partial y} - \delta y \frac{\partial}{\partial y} \left(\frac{d\mathcal{E}_y}{dt} \right). \quad (3.190)$$

Consider the right-hand side of the equation as being a forcing term for the left-hand side. If $\omega_c^{-1} \partial \mathcal{E}_y / \partial y < 1$, then the left-hand side is a simple harmonic oscillator equation in the variable $\delta \dot{y}$. However, if $\omega_c^{-1} \partial \mathcal{E}_y / \partial y$ exceeds unity, then the left-hand side becomes an equation with solutions that grow exponentially in time. Thus, if two particles are initially separated by the infinitesimal distance

δy and if $\omega_c^{-1} \partial \mathcal{E}_y / \partial y < 1$, the separation distance between the two particles will undergo harmonic oscillations, but if $\omega_c^{-1} \partial \mathcal{E}_y / \partial y > 1$ the separation distance will exponentially diverge with time. It is seen that α corresponds to $\omega_c^{-1} \partial \mathcal{E}_y / \partial y$ for a sinusoidal wave. Exponential growth of the separation distance between two particles that are initially arbitrarily close together is called stochastic behavior.

3.8 Particle motion in small-amplitude oscillatory fields

Suppose a small-amplitude electromagnetic field exists in a plasma that in addition has a large, uniform, steady-state magnetic field and no steady-state electric field. The fields can thus be written as

$$\begin{aligned}\mathbf{E} &= \mathbf{E}_1(\mathbf{x}, t) \\ \mathbf{B} &= \mathbf{B}_0 + \mathbf{B}_1(\mathbf{x}, t),\end{aligned}\tag{3.191}$$

where the subscript 1 denotes the small-amplitude oscillatory quantities and the subscript 0 denotes large, uniform equilibrium quantities. A typical particle in this plasma will develop an oscillatory motion

$$\mathbf{x}(t) = \langle \mathbf{x}(t) \rangle + \delta \mathbf{x}(t),\tag{3.192}$$

where $\langle \mathbf{x}(t) \rangle$ is the particle's time-averaged position and $\delta \mathbf{x}(t)$ is the instantaneous deviation from this average position. If the amplitudes of $\mathbf{E}_1(\mathbf{x}, t)$ and $\mathbf{B}_1(\mathbf{x}, t)$ are sufficiently small, then the fields at the particle position can be approximated as

$$\begin{aligned}\mathbf{E}(\langle \mathbf{x}(t) \rangle + \delta \mathbf{x}(t), t) &\simeq \mathbf{E}_1(\langle \mathbf{x}(t) \rangle, t) \\ \mathbf{B}(\langle \mathbf{x}(t) \rangle + \delta \mathbf{x}(t), t) &\simeq \mathbf{B}_0 + \mathbf{B}_1(\langle \mathbf{x}(t) \rangle, t).\end{aligned}\tag{3.193}$$

This is the opposite limit from what was considered in Section 3.7.3. The Lorentz equation reduces in this small-amplitude limit to

$$m \frac{d\mathbf{v}}{dt} = q [\mathbf{E}_1(\langle \mathbf{x} \rangle, t) + \mathbf{v} \times (\mathbf{B}_0 + \mathbf{B}_1(\langle \mathbf{x}(t) \rangle, t))].\tag{3.194}$$

Since the oscillatory fields are small, the resulting particle velocity will also be small (unless there is a resonant response as would happen at the cyclotron frequency). If the particle velocity is small, then the term $\mathbf{v} \times \mathbf{B}_1(\mathbf{x}, t)$ is of second-order smallness, whereas $\mathbf{E}_1$ and $\mathbf{v} \times \mathbf{B}_0$ are of first-order smallness. The $\mathbf{v} \times \mathbf{B}_1(\mathbf{x}, t)$ term is thus insignificant compared to the other two terms on the right-hand side and therefore can be discarded so that the Lorentz equation reduces to

$$m \frac{d\mathbf{v}}{dt} = q [\mathbf{E}_1(\langle \mathbf{x} \rangle, t) + \mathbf{v} \times \mathbf{B}_0],\tag{3.195}$$

a linear differential equation for $\mathbf{v}$. Since $\delta\mathbf{x}$ is assumed to be so small that it can be ignored, the average brackets will be omitted from now on and the first order electric field will simply be written as $\mathbf{E}_1(\mathbf{x}, t)$, where $\mathbf{x}$ can be interpreted as being either the actual or the average position of the particle.

The oscillatory electric field can be decomposed into Fourier modes, each having time dependence $\sim \exp(-i\omega t)$ and, since Eq. (3.195) is linear, the particle response to a field $\mathbf{E}_1(\mathbf{x}, t)$ is just the linear superposition of its response to each Fourier mode. Thus, it is appropriate to consider motion in a single Fourier mode of the electric field, say

$$\mathbf{E}_1(\mathbf{x}, t) = \tilde{\mathbf{E}}(\mathbf{x}, \omega) \exp(-i\omega t). \quad (3.196)$$

If initial conditions are ignored for now, the particle motion can be found by simply assuming that the particle velocity also has the time dependence $\exp(-i\omega t)$, in which case the Lorentz equation becomes

$$-i\omega m \mathbf{v} = q [\tilde{\mathbf{E}}(\mathbf{x}) + \mathbf{v} \times \mathbf{B}_0], \quad (3.197)$$

where a factor $\exp(-i\omega t)$ is implicitly assumed for all terms and also an ω argument is implicitly assumed for $\tilde{\mathbf{E}}$. Equation (3.197) is a vector equation of the form

$$\mathbf{v} + \mathbf{v} \times \mathbf{A} = \mathbf{C}, \quad (3.198)$$

where

$$\begin{aligned} \mathbf{A} &= \frac{\omega_c}{i\omega} \hat{z}, \\ \omega_c &= \frac{qB_0}{m}, \\ \mathbf{C} &= \frac{iq}{\omega m} \tilde{\mathbf{E}}(\mathbf{x}), \end{aligned} \quad (3.199)$$

and the z axis has been chosen to be in the direction of $\mathbf{B}_0$. Equation (3.198) can be solved for $\mathbf{v}$ by first dotting with $\mathbf{A}$ to obtain

$$\mathbf{A} \cdot \mathbf{v} = \mathbf{C} \cdot \mathbf{A} \quad (3.200)$$

and then crossing with $\mathbf{A}$ to obtain

$$\mathbf{v} \times \mathbf{A} + \mathbf{A} \mathbf{A} \cdot \mathbf{v} - \mathbf{v} A^2 = \mathbf{C} \times \mathbf{A}. \quad (3.201)$$

Substituting for $\mathbf{A} \cdot \mathbf{v}$ using Eq. (3.200) and for $\mathbf{v} \times \mathbf{A}$ using Eq. (3.198) gives

$$\mathbf{v} = \frac{\mathbf{C} + \mathbf{A} \mathbf{A} \cdot \mathbf{C} - \mathbf{C} \times \mathbf{A}}{1 + A^2} = C_{\parallel} \hat{z} + \frac{\mathbf{C}_{\perp}}{1 + A^2} + \frac{\mathbf{A} \times \mathbf{C}}{1 + A^2}, \quad (3.202)$$

where $\mathbf{C}$ has been split into parallel and perpendicular parts relative to $\mathbf{B}_0$ and $\mathbf{A} \cdot \mathbf{C} = A^2 C_{\parallel} \hat{\mathbf{z}}$ has been used. On substituting for $\mathbf{A}$ and $\mathbf{C}$ this becomes

$$\mathbf{v} = \frac{iq}{\omega m} \left[\tilde{E}_{\parallel}(\mathbf{x}) \hat{\mathbf{z}} + \frac{\tilde{\mathbf{E}}_{\perp}(\mathbf{x})}{1 - \omega_c^2/\omega^2} - \frac{i\omega_c}{\omega} \frac{\hat{\mathbf{z}} \times \tilde{\mathbf{E}}(\mathbf{x})}{1 - \omega_c^2/\omega^2} \right] e^{-i\omega t}. \quad (3.203)$$

The third term on the right-hand side is a generalization of the $\mathbf{E} \times \mathbf{B}$ drift, since for $\omega \ll \omega_c$ this term reduces to the $\mathbf{E} \times \mathbf{B}$ drift. Similarly, the middle term on the right-hand side is a generalization of the polarization drift, since for $\omega \ll \omega_c$ this term reduces to the polarization drift. The first term on the right-hand side, the parallel quiver velocity, does not involve the magnetic field B_0 . This non-dependence on magnetic field is to be expected because no magnetic force results from motion parallel to a magnetic field. In fact, if the magnetic field were zero, then the second term would add to the first and the third term would vanish, giving a three-dimensional unmagnetized quiver velocity $\mathbf{v} = iq\tilde{\mathbf{E}}(\mathbf{x})/\omega m$.

If the electric field is in addition decomposed into spatial Fourier modes with dependence $\sim \exp(i\mathbf{k} \cdot \mathbf{x})$, then the velocity for a typical mode will be

$$\mathbf{v}(\mathbf{x}, t) = \frac{iq}{\omega m} \left[\tilde{E}_{\parallel} \hat{\mathbf{z}} + \frac{\tilde{\mathbf{E}}_{\perp}}{1 - \omega_c^2/\omega^2} - \frac{i\omega_c}{\omega} \frac{\hat{\mathbf{z}} \times \tilde{\mathbf{E}}}{1 - \omega_c^2/\omega^2} \right] e^{i\mathbf{k} \cdot \mathbf{x} - i\omega t}. \quad (3.204)$$

The convention of a negative coefficient for ω and a positive coefficient for $\mathbf{k}$ has been adopted to give waves propagating in the positive $\mathbf{x}$ direction. Equation (3.204) will later be used as the starting point for calculating wave-generated plasma currents.

3.9 Wave-particle energy transfer

3.9.1 "Average velocity"

Anyone who has experienced delay in a traffic jam knows that it is usually impossible to make up for the lost time by going faster after escaping from the traffic jam. To see why, define α as the fraction of the total trip length in the traffic jam, v_s as the slow (traffic jam) speed, and v_f as the fast speed (out of traffic jam). It is tempting, but wrong, to say that the average velocity is $(1 - \alpha)v_f + \alpha v_s$ because

$$\text{average velocity of a trip} = \frac{\text{total distance}}{\text{total time}}. \quad (3.205)$$

Since the fast-portion duration is $t_f = (1 - \alpha)L/v_f$ while the slow-portion duration is $t_s = \alpha L/v_s$, the average velocity of the complete trip is

$$v_{avg} = \frac{L}{(1 - \alpha)L/v_f + \alpha L/v_s} = \frac{1}{(1 - \alpha)/v_f + \alpha/v_s}. \quad (3.206)$$

Thus, if $v_s \ll v_f$ then $v_{avg} \simeq v_s/\alpha$. The average velocity is almost entirely determined by the slow-portion velocity and as verified by sad experience in traffic jams, once v_s becomes infinitesimal, it is impossible to boost v_{avg} by increasing v_f .

3.9.2 Motion of particles in a sawtooth potential

The exact motion of a particle in a sinusoidal potential can be solved using elliptic integrals, but the obtained solution is implicit, i.e., the solution is expressed in the form of time as a function of position. While exact, the implicit nature of this solution obscures the essential physics. In order to shed some light on the underlying physics, we will first consider particle motion in the somewhat artificial situation of the periodic sawtooth-shaped potential shown in Fig. 3.18 and then later will consider particle motion in a more realistic, but harder to analyze, sinusoidal potential.

A particle in the downward-sloping portion of the sawtooth potential experiences a constant acceleration $+a$ and when in the upward portion experiences a constant acceleration $-a$. The goal here is to determine the average velocity of a group of particles injected with an initial velocity v_0 into the system. Care is required when using the word “average” because this word has two meanings depending on whether one is referring to the average velocity of a single particle or the average velocity of a group of particles. The average velocity of a single particle is defined by Eq. (3.205) whereas the average velocity of a group of particles is defined as the sum of the velocities of all the particles in the group divided by the number of particles in the group.

The average velocity of any given individual particle depends on where the particle was injected. Consider the four particles denoted as A , B , C , and D in Fig. 3.18 as representatives of the various possibilities for injection location. Particle A is injected at a potential maximum, particle C is injected at a potential minimum, particle B is injected half-way on the downslope, and particle D is injected half-way on the upslope.

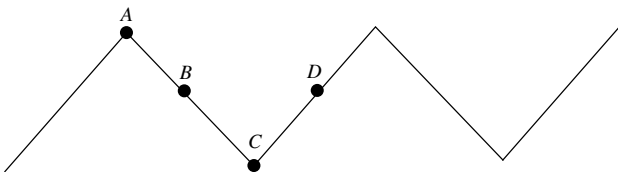


Fig. 3.18 Initial positions of particles A , B , C , and D . All are injected with same initial velocity v_0 moving to the right.

The average velocity for each of these four representative particles will now be evaluated:

Particle *A* – Let the distance between maximum and minimum potential be d . Let $x = 0$ be the location of the minimum so the injection point is at $x = -d$. Thus, the trajectory on the downslope is

$$x(t) = -d + v_0 t + at^2/2 \quad (3.207)$$

and the time for particle *A* to go from its injection point to the potential minimum is found by setting $x(t) = 0$ giving

$$t_{down}^A = \frac{v_0}{a} \left(-1 + \sqrt{1 + 2\delta} \right), \quad (3.208)$$

where $\delta = ad/v_0^2$ is the normalized acceleration. When particle *A* reaches the next potential peak, it again has velocity v_0 and if the time and space origins are reset to be at the new peak, the trajectory will be

$$x(t) = v_0 t - at^2/2. \quad (3.209)$$

The negative time when the particle is at the preceding potential minimum is found from

$$-d = v_0 t - at^2/2. \quad (3.210)$$

Solving for this negative time and then calculating the time increment to go from the minimum to the maximum shows that this time is the same as going from the maximum to the minimum, i.e., $t_{down}^A = t_{up}^A$. Thus, the average velocity for particle *A* is

$$v_{avg}^A = \frac{da/v_0}{-1 + \sqrt{1 + 2\delta}}. \quad (3.211)$$

The average velocity of particle *A* is thus always *faster* than its injection velocity.

Particle *C* – Now let $x = 0$ be the location of maximum potential and $x = -d$ be the point of injection so the particle trajectory is

$$x(t) = -d + v_0 t - at^2/2 \quad (3.212)$$

and the time to get to $x = 0$ is

$$t_{up}^C = \frac{v_0}{a} \left(1 - \sqrt{1 - 2\delta} \right). \quad (3.213)$$

From symmetry it is seen that the time to go from the maximum to the minimum will be the same so the average velocity will be

$$v_{avg}^C = \frac{ad/v_0}{1 - \sqrt{1 - 2\delta}}. \quad (3.214)$$

Because particle *C* is always on a potential hill relative to its injection position, its average velocity is always *slower* than its injection velocity.

Particles *B* and *D* – Particle *B* can be considered as first traveling in a potential well and then in a potential hill, while the reverse is the case for particle *D*. For the potential well portion, the forces are the same as for particle *A*, but the distances are half as much, so the time to traverse the potential well portion is

$$t_{well} = \frac{2v_0}{a} (-1 + \sqrt{1 + \delta}). \quad (3.215)$$

Similarly, the time required to traverse the potential hill portion will be

$$t_{hill} = \frac{2v_0}{a} (1 - \sqrt{1 - \delta}) \quad (3.216)$$

so the average velocity for particles *B* and *D* will be

$$v_{avg}^{B,D} = \frac{ad/v_0}{\sqrt{1 + \delta} - \sqrt{1 - \delta}}. \quad (3.217)$$

These particles move *slower* than the injection velocity, but the effect is second order in δ .

The average velocity of the four particles will be

$$\begin{aligned} v_{avg} &= \frac{1}{4} (v_{avg}^A + v_{avg}^B + v_{avg}^C + v_{avg}^D) \\ &= \frac{ad}{4v_0} \left[\frac{1}{-1 + \sqrt{1 + 2\delta}} + \frac{1}{1 - \sqrt{1 - 2\delta}} + \frac{2}{\sqrt{1 + \delta} - \sqrt{1 - \delta}} \right]. \end{aligned} \quad (3.218)$$

If δ is small this expression can be approximated as

$$\begin{aligned} v_{avg} &\simeq \frac{ad}{4\delta v_0} \left[\frac{1}{1 - \delta/2 + \delta^2/2} + \frac{1}{1 + \delta/2 + \delta^2/2} + \frac{2}{1 + \delta^2/8} \right] \\ &= \frac{ad}{2\delta v_0} \left[\frac{1 + \delta^2/2}{1 + 3\delta^2/4} + \frac{1}{1 + \delta^2/8} \right] \\ &\simeq v_0 [1 - 3\delta^2/16] \end{aligned} \quad (3.219)$$

so that the average velocity of the four representative particles is *smaller* than the injection velocity. This effect is second order in δ and means that a group of particles injected at random locations with identical velocities into a sawtooth periodic potential will, on average, be slowed down.

3.9.3 *Slowing down, energy conservation, and average velocity*

The sawtooth potential analysis above shows that it is necessary to be very careful about what is meant by energy and average velocity. Each particle individually conserves energy and regains its injection velocity when it returns to the phase at which it was injected. However, the average velocities of the particles are not the same as the injection velocities. Particle *A* has an average velocity higher than its injection velocity whereas particles *B*, *C*, and *D* have average velocities smaller than their respective injection velocities. The average velocity of all the particles is less than the injection velocity so that the average kinetic energy of the particles is reduced relative to the injection kinetic energy. Thus the average velocity of a group of particles slows down in a periodic potential, yet paradoxically individual particles do not lose energy. The energy that appears to be missing is contained in the instantaneous potential energy of the individual particles.

3.9.4 *Wave-particle energy transfer in a sinusoidal wave*

The calculation will now be redone for the physically more relevant situation where a group of particles interacts with a sinusoidal wave. As a prerequisite for doing this calculation it must first be recognized that two distinct classes of particles exist, namely those that are trapped in the wave and those that are not. The trajectories of trapped particles differ in a substantive way from those of untrapped particles, but for low-amplitude waves the number of trapped particles is so small as to be of no consequence. It will therefore be assumed that the wave amplitude is sufficiently small that the trapped particles can be ignored.

Particle energy is conserved in the wave frame but not in the lab frame because the particle Hamiltonian is time-independent in the wave frame but not in the lab frame. Since each additional conserved quantity reduces the number of equations to be solved, it is advantageous to calculate the particle dynamics in the wave frame, and then transform back to the lab frame.

The analysis in Section 3.9.2 of particle motion in a sawtooth potential showed that randomly phased groups of particles have their average velocity slowed down, i.e., the average velocity of the group tends towards zero as observed in the frame of the sawtooth potential. If the sawtooth potential were moving with respect to the lab frame, the sawtooth potential would appear as a propagating wave in the lab frame. A lab-frame observer would see the particle velocities tending to come to rest in the sawtooth frame, i.e., the lab-frame average of the particle velocities would tend to converge towards the velocity with which the sawtooth frame moves in the lab frame.

The quantitative motion of a particle in a one-dimensional wave potential $\phi(x, t) = \phi_0 \cos(kx - \omega t)$ will now be analyzed in some detail. This situation

corresponds to a particle being acted on by a wave traveling in the positive x direction with phase velocity ω/k . It is assumed that there is no magnetic field so the equation of motion is simply

$$\frac{dv}{dt} = \frac{qk\phi_0}{m} \sin(kx - \omega t). \quad (3.220)$$

At $t = 0$ the particle's position is $x = x_0$ and its velocity is $v = v_0$. The wave phase at the particle location is defined to be $\psi = kx - \omega t$. This is a more convenient variable than x and so the differential equation for x will be transformed into a corresponding differential equation for ψ . Using ψ as the dependent variable corresponds to transforming to the wave frame, i.e., the frame moving with the phase velocity ω/k , and makes it possible to take advantage of the wave-frame energy being a constant of the motion. The equations are less cluttered with minus signs if a slightly modified phase variable $\theta = kx - \omega t - \pi$ is used.

The respective first and second derivatives of θ are

$$\frac{d\theta}{dt} = kv - \omega \quad (3.221)$$

and

$$\frac{d^2\theta}{dt^2} = k \frac{dv}{dt}. \quad (3.222)$$

Substitution of Eq. (3.222) into Eq. (3.220) gives

$$\frac{d^2\theta}{dt^2} + \frac{k^2 q\phi_0}{m} \sin \theta = 0. \quad (3.223)$$

By defining the *bounce* frequency

$$\omega_b^2 = \frac{k^2 q\phi_0}{m} \quad (3.224)$$

and the dimensionless bounce-normalized time

$$\tau = \omega_b t, \quad (3.225)$$

Eq. (3.223) reduces to the pendulum-like equation

$$\frac{d^2\theta}{d\tau^2} + \sin \theta = 0. \quad (3.226)$$

Upon multiplying by the integrating factor $2d\theta/d\tau$, Eq. (3.226) becomes

$$\frac{d}{d\tau} \left[\left(\frac{d\theta}{d\tau} \right)^2 - 2 \cos \theta \right] = 0. \quad (3.227)$$

This integrates to give

$$\left(\frac{d\theta}{d\tau}\right)^2 - 2\cos\theta = \lambda = \text{const.} \quad (3.228)$$

which indicates the expected energy conservation in the wave frame. The value of λ is determined by two initial conditions, namely the wave-frame injection velocity

$$\left(\frac{d\theta}{d\tau}\right)_{\tau=0} = \frac{1}{\omega_b} \left(\frac{d\theta}{dt}\right)_{t=0} = \frac{kv_0 - \omega}{\omega_b} \equiv \alpha \quad (3.229)$$

and the wave-frame injection phase

$$\theta_{\tau=0} = kx_0 - \pi \equiv \theta_0. \quad (3.230)$$

Inserting these initial values in the left-hand side of Eq. (3.228) gives

$$\lambda = \alpha^2 - 2\cos\theta_0. \quad (3.231)$$

Except for a constant factor,

- λ is the total wave-frame energy,
- $(d\theta/d\tau)^2$ is the wave-frame kinetic energy,
- $-2\cos\theta$ is the wave-frame potential energy.

If $-2 < \lambda < 2$, then the particle is trapped in a specific wave trough and oscillates back and forth in this trough. However, if $\lambda > 2$, the particle is untrapped and travels continuously in the same direction, speeding up when traversing a potential valley and slowing down when traversing a potential hill.

Attention will now be restricted to untrapped particles with kinetic energy greatly exceeding potential energy. For these particles

$$\alpha^2 \gg 2, \quad (3.232)$$

which corresponds to considering small-amplitude waves, since $\alpha \sim \omega_b^{-1}$ and $\omega_b \sim \sqrt{\phi_0}$.

We wish to determine how these untrapped particles exchange energy with the wave. To accomplish this, the lab-frame kinetic energy must be expressed in terms of wave-frame quantities. From Eqs. (3.221) and (3.225) the lab-frame velocity is

$$v = \frac{1}{k} \left(\omega + \frac{d\theta}{dt} \right) = \frac{\omega_b}{k} \left(\frac{\omega}{\omega_b} + \frac{d\theta}{d\tau} \right) \quad (3.233)$$

so that the lab-frame kinetic energy can be expressed as

$$W = \frac{1}{2}mv^2 = \frac{m\omega_b^2}{2k^2} \left[\left(\frac{\omega}{\omega_b} \right)^2 + 2 \frac{\omega}{\omega_b} \frac{d\theta}{d\tau} + \left(\frac{d\theta}{d\tau} \right)^2 \right]. \quad (3.234)$$

Substituting for $(d\theta/d\tau)^2$ using Eq. (3.228) gives

$$W = \frac{m\omega_b^2}{2k^2} \left[\left(\frac{\omega}{\omega_b} \right)^2 + 2 \frac{\omega}{\omega_b} \frac{d\theta}{d\tau} + \lambda + 2 \cos \theta \right]. \quad (3.235)$$

Since wave-particle energy transfer is of interest, attention is now focused on the *changes* in the lab-frame particle kinetic energy and so we consider

$$\begin{aligned} \frac{dW}{dt} &= \frac{m\omega_b^3}{k^2} \left[\frac{\omega}{\omega_b} \frac{d^2\theta}{d\tau^2} - \frac{d\theta}{d\tau} \sin \theta \right] \\ &= -\frac{m\omega_b^3}{k^2} \sin \theta \left[\frac{\omega}{\omega_b} + \frac{d\theta}{d\tau} \right], \end{aligned} \quad (3.236)$$

where Eq. (3.226) has been used. To proceed further, it is necessary to obtain the time dependence of both $\sin \theta$ and $d\theta/dt$.

Solving Eq. (3.228) for $d\theta/d\tau$ and assuming $\alpha \gg 1$ (corresponding to untrapped particles) gives

$$\begin{aligned} \frac{d\theta}{d\tau} &= \pm \sqrt{\lambda + 2 \cos \theta} \\ &= \pm \sqrt{\alpha^2 + 2(\cos \theta - \cos \theta_0)} \\ &= \alpha \left(1 + \frac{2(\cos \theta - \cos \theta_0)}{\alpha^2} \right)^{1/2} \\ &\simeq \alpha + \frac{\cos \theta - \cos \theta_0}{\alpha}. \end{aligned} \quad (3.237)$$

This expression is valid for both positive and negative α , i.e., for particles going in either direction in the wave frame. The first term in the last line of Eq. (3.237) gives the velocity the particle would have if there were *no* wave (unperturbed orbit) while the second term gives the *perturbation* due to the small-amplitude wave. The particle orbit $\theta(\tau)$ is now solved iteratively. To lowest order (i.e., dropping terms of order α^{-2}) the particle velocity is

$$\frac{d\theta}{d\tau} = \alpha \quad (3.238)$$

and so the rate at which energy is transferred from the wave to the particles is

$$\begin{aligned} \frac{dW}{dt} &= -\frac{m\omega_b^3}{k^2} \sin \theta \left[\frac{\omega}{\omega_b} + \alpha \right] \\ &\simeq -\frac{m\omega_b^2 v_0}{k} \sin \theta. \end{aligned} \quad (3.239)$$

Integration of Eq. (3.238) gives the unperturbed orbit solution

$$\theta(\tau) = \theta_0 + \alpha\tau. \quad (3.240)$$

This first approximation is then substituted back into Eq. (3.237) to get the corrected form

$$\frac{d\theta}{d\tau} = \alpha + \frac{\cos(\theta_0 + \alpha\tau) - \cos \theta_0}{\alpha}, \quad (3.241)$$

which may be integrated to give the corrected phase

$$\theta(\tau) = \theta_0 + \alpha\tau + \frac{\sin(\theta_0 + \alpha\tau) - \sin \theta_0}{\alpha^2} - \frac{\tau}{\alpha} \cos \theta_0. \quad (3.242)$$

From Eq. (3.242) we may write

$$\sin \theta = \sin[(\theta_0 + \alpha\tau) + \Delta(\tau)], \quad (3.243)$$

where

$$\Delta(\tau) = \frac{\sin(\theta_0 + \alpha\tau) - \sin \theta_0}{\alpha^2} - \frac{\tau}{\alpha} \cos \theta_0 \quad (3.244)$$

is the “perturbed-orbit” correction to the phase. If consideration is restricted to times where $\tau \ll |\alpha|$, the phase correction $\Delta(\tau)$ will be small. This restriction corresponds to

$$(\omega_b t)^2 \ll |kv_0 - \omega|t, \quad (3.245)$$

which means that the number of wave peaks the particle passes greatly exceeds the number of bounce times. Since bounce frequency is proportional to wave amplitude, this condition will be satisfied for all finite times for an infinitesimal amplitude wave. Because Δ is assumed to be small, Eq. (3.243) may be expanded as

$$\sin \theta = \sin(\theta_0 + \alpha\tau) \cos \Delta + \sin \Delta \cos(\theta_0 + \alpha\tau) \simeq \sin(\theta_0 + \alpha\tau) + \Delta \cos(\theta_0 + \alpha\tau) \quad (3.246)$$

so that Eq. (3.239) becomes

$$\frac{dW}{dt} = -\frac{m\omega_b^2 v_0}{k} [\sin(\theta_0 + \alpha\tau) + \Delta \cos(\theta_0 + \alpha\tau)]. \quad (3.247)$$

The wave-to-particle energy transfer rate depends on the particle initial position. This is analogous to the earlier sawtooth potential analysis where it was shown that whether a particle gains or loses average velocity depends on its injection phase. It is now assumed that there exist many particles with *evenly spaced* initial positions and then an averaging will be performed over all these particles, which

corresponds to averaging over all initial injection phases. Denoting such averaging by $\langle \rangle$ gives

$$\begin{aligned} \left\langle \frac{dW}{dt} \right\rangle &= -\frac{m\omega_b^2 v_0}{k} \langle \Delta \cos(\theta_0 + \alpha\tau) \rangle \\ &= -\frac{m\omega_b^2 v_0}{k} \left\langle \left[\frac{\sin(\theta_0 + \alpha\tau) - \sin \theta_0}{\alpha^2} - \frac{\tau}{\alpha} \cos \theta_0 \right] \cos(\theta_0 + \alpha\tau) \right\rangle. \end{aligned} \quad (3.248)$$

Using the identities

$$\begin{aligned} \langle \sin(\theta_0 + \alpha\tau) \cos(\theta_0 + \alpha\tau) \rangle &= 0 \\ \langle \sin \theta_0 \cos(\theta_0 + \alpha\tau) \rangle &= -\frac{1}{2} \sin \alpha\tau \\ \langle \cos \theta_0 \cos(\theta_0 + \alpha\tau) \rangle &= \frac{1}{2} \cos \alpha\tau \end{aligned} \quad (3.249)$$

the wave-to-particle energy transfer rate becomes

$$\left\langle \frac{dW}{dt} \right\rangle = -\frac{m\omega_b^2 v_0}{2k} \left(\frac{\sin \alpha\tau}{\alpha^2} - \frac{\tau}{\alpha} \cos \alpha\tau \right) = \frac{m\omega_b^2 v_0}{2k} \frac{d}{d\alpha} \left(\frac{\sin \alpha\tau}{\alpha} \right). \quad (3.250)$$

At this point it is recalled that one representation for a delta function is

$$\delta(z) = \lim_{N \rightarrow \infty} \frac{\sin(Nz)}{\pi z} \quad (3.251)$$

so that for $|\alpha\tau| \gg 1$, Eq. (3.250) becomes

$$\left\langle \frac{dW}{dt} \right\rangle = \frac{\pi m\omega_b^2 v_0}{2k} \frac{d}{d\alpha} \delta(\alpha). \quad (3.252)$$

Since $\delta(z)$ has an infinite positive slope just to the left of $z = 0$ and an infinite negative slope just to the right of $z = 0$, the derivative of the delta function consists of a positive spike just to the left of $z = 0$ and a negative spike just to the right of $z = 0$. Furthermore, $\alpha = (kv_0 - \omega)/\omega_b$ is slightly positive for particles moving a little faster than the wave-phase velocity and slightly negative for particles moving a little slower. Thus $\langle dW/dt \rangle$ is large and positive for particles moving slightly slower than the wave, while it is large and negative for particles moving slightly faster. If the number of particles moving slightly slower than the wave equals the number moving slightly faster, the energy gained by the slightly slower particles is equal and opposite to that gained by the slightly faster particles.

However, if the number of slightly slower particles *differs* from the number of slightly faster particles, there will be a net transfer of energy from wave to particles or vice versa. Specifically, if there are more slow particles than fast particles, there will be a transfer of energy to the particles. This energy must come

from the wave and a more complete analysis (cf. Section 5.3) will show that the wave *damps*. The direction of energy transfer depends critically on the slope of the distribution function in the vicinity of $v = \omega/k$, since this slope determines the ratio of slightly faster to slightly slower particles.

We now consider a large number of particles with an initial one-dimensional distribution function $f(v_0)$ and calculate the net wave-to-particle energy transfer rate averaged over all particles. Since $f(v_0)dv_0$ is the probability that a particle had its initial velocity between v_0 and $v_0 + dv_0$, the energy transfer rate averaged over all particles is

$$\begin{aligned} \left\langle \frac{dW_{\text{total}}}{dt} \right\rangle &= \int dv_0 f(v_0) \frac{\pi m \omega_b^2 v_0}{2k} \frac{d}{d\alpha} \delta(\alpha) \\ &= \frac{\pi m \omega_b^3}{2k^2} \int dv_0 f(v_0) v_0 \frac{d}{dv_0} \delta\left(\frac{kv_0 - \omega}{\omega_b}\right) \\ &= \frac{\pi m \omega_b^4}{2k^3} \int dv_0 f(v_0) v_0 \frac{d}{dv_0} \delta\left(v_0 - \frac{\omega}{k}\right) \\ &= -\frac{\pi m \omega_b^4}{2k^3} \left[\frac{d}{dv_0} (f(v_0) v_0) \right]_{v_0=\omega/k}. \end{aligned} \quad (3.253)$$

If the distribution function has the Maxwellian form $f \sim \exp(-v_0^2/v_T^2)$, where v_T is the thermal velocity, and if $\omega/k \gg v_T$ then

$$\begin{aligned} \left[\frac{d}{dv_0} (f(v_0) v_0) \right]_{v_0=\omega/k} &= \left[v_0 \frac{d}{dv_0} (f(v_0)) + f(v_0) \right]_{v_0=\omega/k} \\ &= \left[-2 \frac{v_0^2}{v_T^2} f(v_0) + f(v_0) \right]_{v_0=\omega/k} \end{aligned} \quad (3.254)$$

showing that the derivative of f is the dominant term. Hence, Eq. (3.253) becomes

$$\left\langle \frac{dW_{\text{total}}}{dt} \right\rangle = -\frac{\pi m \omega_b^4 \omega}{2k^4} \left[\frac{d}{dv_0} (f(v_0)) \right]_{v_0=\omega/k}. \quad (3.255)$$

Substituting for the bounce frequency using Eq. (3.224) this becomes

$$\left\langle \frac{dW_{\text{total}}}{dt} \right\rangle = -\frac{\pi m \omega}{2k^2} \left(\frac{qk\phi_0}{m} \right)^2 \left[\frac{d}{dv_0} (f(v_0)) \right]_{v_0=\omega/k}. \quad (3.256)$$

Thus, particles *gain* kinetic energy at the expense of the wave if the distribution function has negative slope in the range $v \sim \omega/k$. This process is called Landau damping and will be examined in Section 5.3 from the wave viewpoint.

3.10 Assignments

1. A charged particle starts from *rest* in combined static fields $\mathbf{E} = E\hat{y}$ and $\mathbf{B} = B\hat{z}$, where $E/B \ll c$ and c is the speed of light. Calculate and plot its exact trajectory (do this both analytically and numerically).
2. Calculate (qualitatively and numerically) the trajectory of a particle starting from rest at $x = 0$, $y = 5a$ in combined E and B fields, where $\mathbf{E} = E_0\hat{x}$ and $\mathbf{B} = \hat{z}B_0y/a$. What happens to μ conservation on the line $y = 0$? Sketch the motion showing both the Larmor motion and the guiding center motion. Explain the particle motion in terms of $\mathbf{E} \times \mathbf{B}$, ∇B , and polarization drifts and note that the particle can never move into the $x < 0$ region because of energy conservation considerations.
3. Calculate the motion of a particle in the steady-state electric field produced by a line charge λ along the z axis and a steady-state magnetic field $\mathbf{B} = B_0\hat{z}$. Obtain an approximate solution using drift theory and also obtain a solution using Hamilton–Lagrange theory. Hint: for the drift theory show that the electric field has the form $\mathbf{E} = \hat{r}\lambda/2\pi r$. Assume that λ is small for approximate solutions.
4. Consider the magnetic field produced by a toroidal coil system; this coil consists of a single wire threading the hole of a torus (donut) N times with the N turns evenly arranged around the circumference of the torus. Use Ampère’s law to show that the magnetic field is in the toroidal direction and has the form $B = \mu_0 NI/2\pi r$, where N is the total number of turns in the coil and I is the current through the turn. What are the drifts for a particle having finite initial velocities both parallel and perpendicular to this toroidal field?
5. Show that of all the standard drifts ($\mathbf{E} \times \mathbf{B}$, ∇B , curvature, polarization) only the polarization drift causes a change in the particle energy. Hint: consider what happens when the following equation is dotted with $\mathbf{v}$:

$$m \frac{d\mathbf{v}}{dt} = \mathbf{F} + \mathbf{v} \times \mathbf{B}.$$

6. Use the numerical Lorentz solver to calculate the motion of a charged particle in a uniform magnetic field $\mathbf{B} = B\hat{z}$ and an electric field given by Eq. (3.178). Compare the motion to the predictions of drift theory ($\mathbf{E} \times \mathbf{B}$, polarization). Describe the motion for cases where $\alpha \ll 1$, $\alpha \simeq 1$, and $\alpha \gg 1$, where $\alpha = mk^2\phi/qB^2$. Describe what happens when α becomes of order unity.
7. A “magnetic mirror” field in cylindrical coordinates r, θ, z can be expressed as $\mathbf{B} = (2\pi)^{-1} \nabla\psi \times \nabla\theta$, where $\psi = B_0\pi r^2(1 + (z/L)^2)$ and L is a characteristic length. Sketch by hand the field line pattern in the r, z plane and write out the components of $\mathbf{B}$. What are appropriate characteristic lengths, times, and velocities for an electron in this configuration? Use $r = (x^2 + y^2)^{1/2}$ and numerically integrate the orbit of an electron starting at $x = 0$, $y = L$, $z = 0$ with initial velocity $v_x = 0$ and initial v_y, v_z of the order of the characteristic velocity (try different values). Simultaneously plot the motion in the $z - y$ plane and in the $x - y$ plane. What interesting phenomena can be observed (e.g., reflection)? Does the electron stay on a constant ψ contour?

8. Consider the motion of a charged particle in the magnetic field

$$\mathbf{B} = \frac{1}{2\pi} \nabla \psi(r, z, t) \times \nabla \theta,$$

where

$$\psi(r, z, t) = B_{\min} \pi r^2 \left[1 + 2\lambda \frac{\zeta^2}{\zeta^4 + 1} \right]$$

and

$$\zeta = \frac{z}{L(t)}.$$

Show by explicit evaluation of the flux derivatives and also by plotting contours of constant flux that this is an example of a magnetic mirror field with minimum axial field $B_{\min}$ when $z = 0$ and maximum axial field $\lambda B_{\min}$ at $z = L(t)$. By making $L(t)$ a slowly decreasing function of time show that the magnetic mirrors slowly move together. Using numerical techniques to integrate the equation of motion, demonstrate Fermi acceleration of a particle when the mirrors move slowly together. Do not forget the electric field associated with the time-changing magnetic field (this electric field is closely related to the time derivative of $\psi(r, z, t)$; use Faraday's law). Plot the velocity-space angle at $z = 0$ for each bounce between mirrors and show that the particle becomes detrapped when this angle decreases below $\theta_{\text{trap}} = \sin^{-1}(\lambda^{-1})$.

9. Consider a point particle bouncing with nominal velocity v between a stationary wall and a second wall that is approaching the first wall with speed u . Calculate the change in speed of the particle after it bounces from the moving wall (hint: do this first in the frame of the moving wall, and then translate back to the lab frame). Calculate τ_b , the time for the particle to make one complete bounce between the walls if the nominal distance between walls is L . Calculate ΔL , the change in L during one complete bounce and show that if $u \ll v$, then Lv is a conserved quantity. By considering collisionless particles bouncing in a cube that is slowly shrinking self-similarly in three dimensions show that $PV^{5/3}$ is constant where $P = nkT$, n is the density of the particles, and T is the average kinetic energy of the particles. What happens if the shrinking is not self-similar (hint: consider the effect of collisions and see discussion in Bellan (2004a)).
10. Using numerical techniques to integrate the equation of motion, illustrate how a charged particle changes from being non-axis-encircling to axis-encircling when a magnetic field $\mathbf{B} = (2\pi)^{-1} \nabla \psi(r, z, t) \times \nabla \theta$ reverses polarity at $t = 0$. For simplicity use $\psi = B(t) \pi r^2$, i.e., a uniform magnetic field. To make the solution as general as possible, normalize time to the cyclotron frequency by defining $\tau = \omega_c t$, and set $B(\tau) = \tanh \tau$ to represent a polarity-reversing field. Normalize lengths to some reference length L and normalize velocities to $\omega_c L$. Show that the canonical angular momentum is conserved. Hint: do not forget to include the inductive electric field associated with the time-dependent magnetic field.

11. Consider a cusp magnetic field given by $\mathbf{B} = (2\pi)^{-1} \nabla\psi(r, z) \times \nabla\theta$, where the flux function

$$\psi(r, z) = B\pi r^2 \frac{z}{\sqrt{1 + z^2/a^2}}$$

is antisymmetric in z . Plot the surfaces of constant flux. Using numerical techniques to integrate the equation of motion, demonstrate that a particle incident at $z \ll -a$ and $r = r_0$ with incident velocity $\mathbf{v} = v_{z0}\hat{z}$ will reflect from the cusp if $v_{z0} < r_0\omega_c$, where $\omega_c = qB/m$.

12. Consider the motion of a charged particle starting from rest in a simple one-dimensional electrostatic wave field,

$$m \frac{d^2x}{dt^2} = -q\nabla\phi(x, t),$$

where $\phi(x, t) = \bar{\phi} \cos(kx - \omega t)$. How large does $\bar{\phi}$ have to be to give trapping of particles that start from rest? Demonstrate this trapping threshold numerically.

13. Prove Equation (3.219).

14. Prove that

$$\delta(z) = \lim_{N \rightarrow \infty} \frac{\sin(Nz)}{\pi z}$$

is a valid representation for the delta function.

15. As sketched in Fig. 3.19, a current loop (radius r , current I) is located in the $x-y$ plane; the loop's axis defines the z axis of the coordinate system, so that the center of the loop is at the origin. The loop is immersed in a non-uniform magnetic field $\mathbf{B}$ produced by external coils and oriented so that the magnetic-field lines converge symmetrically about the z axis. The current I is small and does not significantly modify $\mathbf{B}$. Consider the following three circles: the current loop, a circle of radius b coaxial with the loop but with center at $z = -L/2$, and a circle of radius a with

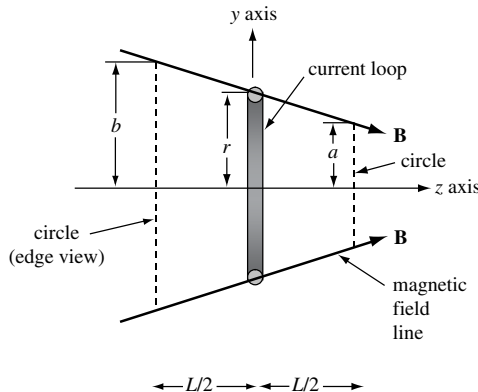


Fig. 3.19 Non-uniform magnetic field acting on a current loop.

center at $z = +L/2$. The radii a and b are chosen so as to intercept the field lines that intercept the current loop (see figure). Assume the figure is somewhat exaggerated so that B_z is approximately uniform over each of the three circular surfaces and so one may ignore the radial dependence of B_z and therefore express $B_z = B_z(z)$.

- (a) Note that $r = (a + b)/2$. What is the force (magnitude and direction) on the current loop expressed in terms of I , $B_z(0)$, a , b , and L only? (Hint: use the field line slope to give a relationship between B_r and B_z at the loop radius.)
- (b) For each of the circles and the current loop, express the magnetic flux enclosed in terms of B_z at the respective entity and the radius of the entity. What is the relationship between the B_z 's at these three entities?
- (c) By combining the results of parts (a) and (b) above and taking the limit $L \rightarrow 0$, show that the force on the loop can be expressed in terms of a derivative of B_z .